Publications by Model: RAPID

Total Publications: 136 (sorted newest to oldest)

1. Impacts of Different Satellite-Based Precipitation Signature Errors on Hydrological Modeling Performance Across China

Authors: Chiyuan Miao (Beijing Normal University); Jiaojiao Gou (Beijing Normal University); Jinlong Hu (Beijing Normal University); Qingyun Duan (Hohai University)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024ef004954

URL: https://doi.org/10.1029/2024ef004954

Publication Date: 2024-11-23

Volume/Issue: Volume 12, Issue 11

References: 77

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

Open Access: N/A

Abstract: The quasi-global availability of satellite-based precipitation products (SPPs) holds significant potential for improving hydrological modeling skill. However, limited knowledge exists concerning the impacts of different SPP error type on hydrological modeling skill and their sensitivity across different climate zones. In this study, forcing data sets from 10 SPPs were collected to drive hydrological models during the period 2001-2018 for 366 catchments across China. Here, we analyze the impact of the SPP errors associated with different precipitation intensities (light, moderate, and heavy) and different precipitation signatures (magnitude, variance, and occurrence) on the performance of hydrological simulations, and rank the sensitivities of SPPs errors for four major Köppen-Geiger climate zones. The results show that heavy precipitation in SPPs is generally associated with higher errors than light and moderate precipitation when compared to gauge-based precipitation observations, but hydrological model skill is more sensitive to errors from moderate precipitation than from heavy precipitation. The probability of moderate precipitation detection was identified as the most sensitive metric in determining hydrological model performance, with sensitivities of 0.58, 0.39, 0.59, and 0.47 in the temperate, boreal, arid, and highland climate zones, respectively. The variance error and magnitude error for heavy precipitation from SPPs were also identified as sensitive factors for hydrological modeling in the temperate and arid climate zones, respectively. These findings are crucial for enhancing the understanding of interactions between SPPs uncertainty and hydrological simulations, leading to

improved data accuracy of precipitation forcing and the identification of appropriate SPPs for hydrological simulation in China.

2. State of continental discharge estimation and modelling: challenges and opportunities for Africa

Authors: Komlavi Akpoti (International Water Management Institute); Kirubel Mekonnen (International Water Management Institute); Mansoor Leh (International Water Management Institute); Afua Owusu (International Water Management Institute); Moctar Dembélé (International Water Management Institute); Primrose Tinonetsana (International Water Management Institute); Abdulkarim Seid (International Water Management Institute); Naga Manohar Velpuri (International Water Management Institute)

Journal: Hydrological Sciences Journal

Publisher: Informa UK Limited

DOI: 10.1080/02626667.2024.2402938

URL: https://doi.org/10.1080/02626667.2024.2402938

Publication Date: 2024-10-02

Volume/Issue: Volume 69, Issue 15

References: 140

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Leona M. and Harry B. Helmsley charitable trust

Open Access: N/A

Abstract: N/A

3. Physics-enhanced machine learning models for streamflow discharge forecasting

Authors: Ying Zhao (University of Michigan); Mayank Chadha (University of California San Diego); Dakota Barthlow (c Hottinger Bruel & Kjaer Solutions LLC, Southfield, MI, 48076, USA); Elissa Yeates (d Coastal and Hydraulics Laboratory, Engineer Research and Development Center, US Army Corps of Engineers, Vicksburg, MS, 39180, USA); Charles J. Mcknight (d Coastal and Hydraulics Laboratory, Engineer Research and Development Center, US Army Corps of Engineers, Vicksburg, MS, 39180, USA); Natalie P. Memarsadeghi (d Coastal and Hydraulics Laboratory, Engineer Research and Development Center, US Army Corps of Engineers, Vicksburg, MS, 39180, USA); Guga Gugaratshan (c Hottinger Bruel & Kjaer Solutions LLC, Southfield, MI, 48076, USA); Michael D. Todd (University of California San Diego); Zhen Hu (University of Michigan)

Journal: Journal of Hydroinformatics

Publisher: IWA Publishing

DOI: 10.2166/hydro.2024.061

URL: https://doi.org/10.2166/hydro.2024.061

Publication Date: 2024-09-23

Volume/Issue: Volume 26, Issue 10

References: 45

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Coastal and Hydraulics Laboratory

Open Access: N/A

Abstract: Accurate river discharge forecasts for short to intermediate time intervals are crucial for decision-making related to flood mitigation, the seamless operation of inland waterways management, and optimal dredging. River routing models that are physics based, such as RAPID ('routing application for parallel computation of discharge') or its variants, are used to forecast river discharge. These physics-based models make numerous assumptions, including linear process modeling, accounting for only adjacent river inflows, and requiring brute force calibration of hydrological input parameters. As a consequence of these assumptions and the missing information that describes the complex dynamics of rivers and their interaction with hydrology and topography, RAPID leads to noisy forecasts that may, at times, substantially deviate from the true gauged values. In this article, we propose hybrid river discharge forecast models that integrate physics-based RAPID simulation model with advanced data-driven machine learning (ML) models. They leverage runoff data of the watershed in the entire basin, consider the physics-based RAPID model, take into account the variability in predictions made by the physics-based model relative to the true gauged discharge values, and are built on state-of-the-art ML models with different complexities. We deploy two different algorithms to build these hybrid models, namely, delta learning and data augmentation. The results of a case study indicate that a hybrid model for discharge predictions outperforms RAPID in terms of overall performance. The prediction accuracy for various rivers in the case study can be improved by a factor of four to seven.

4. Hydrology and Droughts in the Nile: A Review of Key Findings and Implications

Authors: Meklit Berihun Melesse (Department of Civil and Environmental Engineering, Washington State University, Richland, WA 99352, USA); Yonas Demissie (Department of Civil and Environmental Engineering, Washington State University, Richland, WA 99352, USA)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w16172521

URL: https://doi.org/10.3390/w16172521

Publication Date: 2024-09-05

Volume/Issue: Volume 16, Issue 17

References: 141

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Abstract: The Nile Basin has long been the subject of extensive research, reflecting its importance, which spans from its historical role in the development of ancient civilizations to its current significance in supporting rapidly changing socioeconomic conditions of the basin countries. This review synthesizes studies focusing on the past and future climate, hydrologic, and drought outlooks of the basin, and explores the roles played by large-scale atmospheric phenomena and water infrastructure on the basin's climate and hydrology. Overall, the studies underscore the complexity of the Nile hydrological system and the necessity for improved modeling and data integration. This review serves as a guide to areas warranting further research by highlighting the uncertainties and inconsistencies among the different studies. It underscores the interconnectedness of climatic and hydrological processes in the basin and encourages the use of diverse data sources to address the data scarcity issue and ensemble models to reduce modeling uncertainty in future research. By summarizing the data and modeling resources employed in these studies, this review also provides a valuable resource for future modeling efforts to understand and explore of the basin's complex climatic and hydrological dynamics.

5. Extracting an accurate river network: Stream burning re-revisited

Authors: Qiuyang Chen; Simon M. Mudd; Mikael Attal; Steven Hancock

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2024.114333

URL: https://doi.org/10.1016/j.rse.2024.114333

Publication Date: 2024-07-29

Volume/Issue: Volume 312

References: 103

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

6. A Stream-Order Family and Order-Based Parallel River Network Routing Method

Authors: Xi Yang (College of Surveying and Geo-Informatics, North China University of Water Resources and Electric Power, Zhengzhou 450046, China); Chong Wei (College of Surveying and Geo-Informatics, North China University of Water Resources and Electric Power, Zhengzhou 450046, China); Zhiping Li (College of Geosciences and Engineering, North China University of Water Resources and Electric Power, Zhengzhou 450045, China); Heng Yang (Science and Technology Research Institute, China Three Gorges Corporation, Beijing 100038, China); Hui Zheng (Chinese Academy of Sciences)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w16141965

URL: https://doi.org/10.3390/w16141965

Publication Date: 2024-07-11

Volume/Issue: Volume 16, Issue 14

References: 21

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

Open Access: N/A

Abstract: River network routing's significance in reach-level flood forecasting over extensive domains is growing, requiring considerable computational resources for modeling networks comprising thousands to millions of reaches. Parallel computation plays a central role in timely forecasting in such cases. However, the sequentiality of upstream-to-downstream flow paths within river networks poses a significant challenge for parallelization. This study introduces a family of stream orders and an associated order-based parallel routing approach. We assign each reach an order that falls between one more than the maximum order of its upstream reaches and one less than the order of its downstream reach. This strategy enables the parallel simulation of reaches with identical orders while sequentially processing those with different orders, thus maintaining the crucial upstream-to-downstream dynamic. To further enhance parallel scalability, we strategically relax the upstream-to-downstream relationship along the longest flow paths, dividing the network into independent subnetworks and introducing halo reaches to mitigate the impact of inexact inflows. We validate our approach using China's Yangtze River basin, the country's largest river network with 53,600 fully connected reaches. Employing a conceptual parallel execution machine, we demonstrate that our method achieves 80% parallel efficiency with up to 25 processors. By strategically introducing breakpoints, we further enhance scalability, enabling efficient simulations on 77 processors while maintaining 80% efficiency. These results highlight the scalability and efficiency of our methods for large-scale, high-resolution river network modeling within Earth system models. Our study also lays a theoretical groundwork for optimizing stream orders and halo reach placements, crucial for advancing river network modeling.

7. Integrating a Water Tracer Model Into WRF-Hydro for Characterizing the Effect of Lateral Flow in Hydrologic Simulations

Authors: Huancui Hu (Pacific Northwest National Laboratory); L. Ruby Leung (Pacific Northwest National Laboratory); Francina Dominguez (University of Illinois); David Gochis (National Center for Atmospheric Research); Xingyuan Chen (Pacific Northwest National Laboratory); Stephen Good (Oregon State University); Aubrey Dugger (National Center for Atmospheric Research); Laurel Larsen (University of California Berkeley); Michael Barlage (Now at NOAA/Environmental Modeling Center College Park MD USA, National Center for Atmospheric Research)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023wr034938

URL: https://doi.org/10.1029/2023wr034938

Publication Date: 2024-07-08

Volume/Issue: Volume 60, Issue 7

References: 105

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: transit times, lateral flow, WRF-hydro

Funding: Office of Science

Open Access: N/A

Abstract: Most current land models approximate terrestrial hydrological processes as one-dimensional vertical flow, neglecting lateral water movement from ridges to valleys. Such lateral flow is fundamental at catchment scales and becomes crucial for finer-scale land models. To test the effect of incorporating lateral flow toward three-dimensional representations of hydrological processes in the next generation land models, we integrate a water tracer model into the WRF-Hydro framework to track water movement from precipitation to discharge and evapotranspiration. This hydrologic-tracer integrated system allows us to identify the key mechanisms by which lateral flow affects the flow paths and transit times in WRF-Hydro. By comparing modeling experiments with and without lateral routing in two contrasting catchments, we determine the impacts of lateral flow on the transit times of precipitation event-water. Results show that with limited hydrologic connectivity, lateral flow extends the transit times by reducing (increasing) event-water drainage loss (accumulation) in ridges (valleys) and allowing reinfiltration of infiltration-excess flow, which is missing in most land models. On the contrary with high hydrologic connectivity, lateral flow can effectively accelerate the water release to streams and reduce the transit time. However, the transit times are substantially underestimated by the model compared with isotope-derived estimates, indicating model limitations in representing flow paths and transit times. This study provides some insights on the fundamental differences in terrestrial hydrology simulated by land models with and without lateral flow representation.

8. Peak Flow Event Durations in the Mississippi River Basin and Implications for Temporal Sampling of Rivers

Authors: Arnaud Cerbelaud (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sylvain Biancamaria (Institut de Recherche pour le Développement (IRD)); Jeffrey Wade (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Manu Tom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Renato Prata de Moraes Frasson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Denis Blumstein (CNES Centre National D'Études Spatiales Toulouse France, Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl109220

URL: https://doi.org/10.1029/2024gl109220

Publication Date: 2024-06-08

Volume/Issue: Volume 51, Issue 11

References: 53

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: discharge statistics, satellite remote sensing, surface water hydrology, river dynamics, radar altimetry

Funding: Jet Propulsion Laboratory

Open Access: N/A

Abstract: The impact of an episodic river flood is intimately linked to its duration. Yet it is still unclear how often should a river be observed to accurately determine the occurrence and duration of extreme events. Here we assess flow statistics along with peak flow event detection and duration as a function of the discharge sampling period for large tributaries of the Mississippi basin using hourly gages over 2010-2022. Median event durations above high quantiles spatially vary from around 2 days upstream to 30 days downstream. Discharge mean, standard deviation, and quantiles can all be estimated within 2.5% error for sampling periods up to 8 days. A minimum temporal sampling 4x (2x) finer than peak flow event median duration is required to detect 95 +/- 3% (85 +/- 5%) of events and to estimate their duration within 90 +/- 5% (75 +/- 10%) median accuracy. Our findings have direct implications for future satellite missions concerned with capturing flood events.

9. A Flexible Framework for Simulating the Water Balance of Lakes and Reservoirs From Local to Global Scales: mizuRoute-Lake

Authors: Shervan Gharari (Centre for Hydrology University of Saskatchewan Saskatoon SK Canada); Inne Vanderkelen (University of Bern, Vrije Universiteit Brussel); Andrew Tefs (University of Calgary); Naoki Mizukami (National Center for Atmospheric Research); Erik Kluzek (National Center for Atmospheric Research); Tricia Stadnyk (University of Calgary); David Lawrence (National Center for Atmospheric Research); Martyn P. Clark (University of Calgary)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022wr032400

URL: https://doi.org/10.1029/2022wr032400

Publication Date: 2024-05-06

Volume/Issue: Volume 60, Issue 5

References: 69

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: reservoirs, water balance models, lakes, mizuRoute-Lake

Funding: Global Water Futures, Fonds Wetenschappelijk Onderzoek

Open Access: N/A

Abstract: Lakes and reservoirs are an integral part of the terrestrial water cycle. In this work, we present the implementation of different water balance models of lakes and reservoirs into mizuRoute, a vector-based routing model, termed mizuRoute-Lake. As the main advantage of mizuRoute-Lake, users can choose between various parametric models implemented in mizuRoute-Lake. So far, three parametric models of lake and reservoir water balance, namely Hanasaki, HYPE, and Döll are implemented in mizuRoute-Lake. In general, the parametric models relate the outflow from lakes or reservoirs to the storage and various parameters including inflow, demand, volume of storage, etc. Additionally, this flexibility allows users to easily evaluate and compare the effect of various water balance models for a lake without needing to reconfigure the routing model or change the parameters of other lakes or reservoirs in the modeling domain. Users can also use existing data such as historical observations or water management models to specify the behavior of a selected number of lakes and reservoirs within the modeling domain using the data-driven capability of mizuRoute-Lake. We demonstrate the flexibility of mizuRoute-Lake by presenting global, regional, and local scale applications. The development of mizuRoute-Lake paves the way for better integration of water management models, locally measured, and remotely sensed data sets in the context of Earth system modeling.

10. Global patterns in river water storage dependent on residence time

Authors: Elyssa L. Collins; Cédric H. David; Ryan Riggs; George H. Allen; Tamlin M. Pavelsky; Peirong Lin; Ming Pan; Dai Yamazaki; Ross K. Meentemeyer; Georgina M. Sanchez

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-024-01421-5

URL: https://doi.org/10.1038/s41561-024-01421-5

Publication Date: 2024-04-22

Volume/Issue: Volume 17, Issue 5

References: 76

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA | Jet Propulsion Laboratory

Abstract: Accurate assessment of global river flows and stores is critical for informing water management practices, but current estimates of global river flows exhibit substantial spread and estimates of river stores remain sparse. Estimates of river flows and stores are hampered by uncertainties in land runoff, an unobserved quantity that provides water input to rivers. Here we leverage global river flow observations and an ensemble of land surface models to generate a globally gauge-corrected monthly river flow and storage dataset. We estimate a global river storage mean (+/- monthly variability) of 2,246 +/- 505 km3 and a global continental flow of 37,411 +/- 7,816 km3 yr-1. Our global river water storage time series demonstrates that flow wave residence time is a fundamental driver that can double or halve river water stores and their variability. We also reconcile the wide range in previous estimates of monthly variability in global river flows. We identify previously underappreciated freshwater sources to the ocean from the Maritime Continent (Indonesia, Malaysia and Papua New Guinea) amounting to 1.6 times the Congo River and illustrate our capability of detecting severe anthropogenic water withdrawals.

11. A systematic review of Muskingum flood routing techniques

Authors: Aryan Salvati (Department of Arid and Mountainous Regions Reclamation, Faculty of Natural Resources, University of Tehran, Karaj, Iran); Alireza Moghaddam Nia (Department of Arid and Mountainous Regions Reclamation, Faculty of Natural Resources, University of Tehran, Karaj, Iran); Ali Salajegheh (Department of Arid and Mountainous Regions Reclamation, Faculty of Natural Resources, University of Tehran, Karaj, Iran); Ataollah Shirzadi (Department of Rangeland and Watershed Management, Faculty of Natural Resources, University of Kurdistan, Sanandaj, Iran); Himan Shahabi (Departments of Geomorphology, Faculty of Natural Resources, University of Kurdistan, Sanandaj, Iran); Ebrahim Ahmadisharaf (Department of Civil and Environmental Engineering, FAMU-FSU College of Engineering, Tallahassee, FL, USA); Dawei Han (University of Bristol); John J Clague (Department of Earth Sciences, Simon Fraser University, Burnaby, BC, Canada)

Journal: Hydrological Sciences Journal

Publisher: Informa UK Limited

DOI: 10.1080/02626667.2024.2324132

URL: https://doi.org/10.1080/02626667.2024.2324132

Publication Date: 2024-03-06

Volume/Issue: Volume 69, Issue 6

References: 315

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

12. Satellite-based precipitation error propagation in the hydrological modeling chain across China

Authors: Jiaojiao Gou; Chiyuan Miao; Soroosh Sorooshian; Qingyun Duan; Xiaoying Guo; Ting Su

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2024.130906

URL: https://doi.org/10.1016/j.jhydrol.2024.130906

Publication Date: 2024-02-15

Volume/Issue: Volume 632

References: 70

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fundamental Research Funds for the Central Universities, National Natural Science Foundation of China, China Postdoctoral Science Foundation (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

13. Review of Machine Learning Methods for River Flood Routing

Authors: Li Li (Global Engineering Institute for Ultimate Society (GENIUS), Sungkyunkwan University, Suwon 16419, Republic of Korea); Kyung Soo Jun (Graduate School of Water Resources, Sungkyunkwan University, Suwon 16419, Republic of Korea)

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w16020364

URL: https://doi.org/10.3390/w16020364

Publication Date: 2024-01-22

Volume/Issue: Volume 16, Issue 2

References: 86

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Korea Environmental Industry and Technology Institute

Abstract: River flood routing computes changes in the shape of a flood wave over time as it travels downstream along a river. Conventional flood routing models, especially hydrodynamic models, require a high quality and quantity of input data, such as measured hydrologic time series, geometric data, hydraulic structures, and hydrological parameters. Unlike physically based models, machine learning algorithms, which are data-driven models, do not require much knowledge about underlying physical processes and can identify complex nonlinearity between inputs and outputs. Due to their higher performance, lower complexity, and low computation cost, researchers introduced novel machine learning methods as a single application or hybrid application to achieve more accurate and efficient flood routing. This paper reviews the recent application of machine learning methods in river flood routing.

14. Case study of continental-scale hydrologic modeling's ability to predict daily streamflow percentiles for regulatory application

Authors: Joseph L. Gutenson (University of Alabama, Coastal and Hydraulics Laboratory (CHL) U.S. Army Engineer Research and Development Center (ERDC) Vicksburg Mississippi USA); Kent H. Sparrow (Geospatial Research Laboratory (GRL) U.S. Army Engineer Research and Development Center (ERDC) Alexandria Virginia USA); Stephen W. Brown (Coastal and Hydraulics Laboratory (CHL) U.S. Army Engineer Research and Development Center (ERDC) Vicksburg Mississippi USA); Mark D. Wahl (Coastal and Hydraulics Laboratory (CHL) U.S. Army Engineer Research and Development Center (ERDC) Vicksburg Mississippi USA); Kyle B. Gordon (Environmental Laboratory (EL) U.S. Army Engineer Research and Development Center (ERDC) Vicksburg Mississippi USA)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.13189

URL: https://doi.org/10.1111/1752-1688.13189

Publication Date: 2024-01-12

Volume/Issue: Volume 60, Issue 2

References: 56

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Engineer Research and Development Center

License: N/A

Open Access: N/A

Abstract: Regulatory practitioners use hydroclimatic data to provide context to observations typically collected through field site visits and aerial imagery analysis. In the absence of site-specific data, regulatory practitioners must use proxy hydroclimatic data and models to assess a stream's hydroclimatology. One intent of current-generation continental-scale hydrologic models is to provide such hydrologic context to ungaged watersheds. In this study, the ability of two state-of-the-art, operational, continental-scale hydrologic modeling frameworks, the National Water Model and the Group on Earth Observation Global Water Sustainability (GEOGloWS) European Centre for Medium-Range Weather Forecasts (ECMWF) Streamflow Model, to produce daily streamflow percentiles and categorical estimates of the streamflow normalcy was examined. The modeled streamflow percentiles were compared to observed daily streamflow percentiles at four United States Geological Survey stream gages. The model's performance was then compared to a baseline assessment methodology, the Antecedent Precipitation Tool. Results indicated that, when compared to baseline assessment techniques, the accuracy of the National Water Model (NWM) or GEOGloWS ECMWF Streamflow Model was greater than the accuracy of the baseline assessment methodology at four stream gage locations. The NWM performed best at three of the four gages. This work highlighted a novel application of current-generation continental-scale hydrologic models.

15. AltiMaP: altimetry mapping procedure for hydrography data

Authors: Menaka Revel; Xudong Zhou; Prakat Modi; Jean-François Cretaux; Stephane Calmant; Dai Yamazaki

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-16-75-2024

URL: https://doi.org/10.5194/essd-16-75-2024

Publication Date: 2024-01-08

Volume/Issue: Volume 16, Issue 1

References: 63

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science

Open Access: N/A

Abstract: . Satellite altimetry data are useful for monitoring water surface dynamics, evaluating and calibrating hydrodynamic models, and enhancing river-related variables through optimization or assimilation approaches. However, comparing simulated water surface elevations (WSEs) using satellite altimetry data is challenging due to the difficulty of correctly matching the representative locations of satellite altimetry virtual stations (VSs) to the discrete river grids used in hydrodynamic models. In this study, we introduce an automated altimetry mapping procedure (AltiMaP) that allocates VS locations listed in the HydroWeb database to the Multi-Error Removed Improved Terrain Hydrography (MERIT Hydro) river network. Each VS was flagged according to the land cover of the initial pixel allocation, with 10, 20, 30 and 40 representing river channel, land with the nearest single-channel river, land with the nearest multi-channel river and ocean pixels, respectively. Then, each VS was assigned to the nearest MERIT Hydro river reach according to geometric distance. Among the approximately 12 000 allocated VSs, most were categorized as flag 10 (71.7 %). Flags 10 and 20 were mainly located in upstream and midstream reaches, whereas flags 30 and 40 were mainly located downstream. Approximately 0.8 % of VSs showed bias, with considerable elevation differences (>=|15| m) between the mean observed WSE and MERIT digital elevation model. These biased VSs were predominantly observed in narrow rivers at high altitudes. Following VS allocation using AltiMaP, the median root mean square error of simulated WSEs compared to satellite altimetry was 7.86 m. The error rate was improved meaningfully (10.6 %) compared to that obtained using a traditional approach, partly due to bias reduction. Thus, allocating VSs to a river network using the proposed AltiMaP framework improved our comparison of WSEs simulated by the global hydrodynamic model to those obtained by satellite altimetry. The AltiMaP source code (https://doi.org/10.5281/zenodo.7597310, Revel et al., 2023a) and data (https://doi.org/10.4211/hs.632e550deaea46b080bdae986fd19156, Revel et al., 2022) are freely accessible online, and we anticipate that they will be beneficial to the international hydrological community.

16. Reducing Uncertainties of a Chained Hydrologic-Hydraulic Models to Improve Flood Forecasting Using Multi-Source Earth Observation Data

Authors: T. H. Nguyen (Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS),Toulouse Cedex 1,France,31057); S. Ricci (Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS),Toulouse Cedex 1,France,31057); A. Piacentini (Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS),Toulouse Cedex 1,France,31057); Q. Bonassies (Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS),Toulouse Cedex 1,France,31057); R. Rodriquez Suquet (Centre National d'Études Spatiales (CNES),Toulouse Cedex 9,France,31401); S. Peña Luque (Centre National d'Études Spatiales (CNES),Toulouse Cedex 9,France,31401); K. Marlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium

Publisher: IEEE

DOI: 10.1109/igarss52108.2023.10282456

URL: https://doi.org/10.1109/igarss52108.2023.10282456

Publication Date: 2023-10-20

Volume/Issue: N/A

References: 12

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Forcing (mathematics)|Flood forecasting|Hydrological modelling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

17. Hydrological model-based streamflow reconstruction for Indian sub-continental river basins, 1951-2021

Authors: Dipesh Singh Chuphal; Vimal Mishra

Journal: Scientific Data

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41597-023-02618-w

URL: https://doi.org/10.1038/s41597-023-02618-w

Publication Date: 2023-10-18

Volume/Issue: Volume 10, Issue 1

References: 56

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Earth Sciences

Open Access: N/A

Abstract: Streamflow is a vital component of the global water cycle. Long-term streamflow observations are required for water resources planning and management, hydroclimatic extremes analysis, and ecological assessment. However, long-term streamflow observations for the Indian-Subcontinental (ISC) river basins are lacking. Using meteorological observations, state-of-the-art hydrological model, and river routing model, we developed hydrological model-simulated monthly streamflow from 1951-2021 for the ISC river basins. We used high-resolution vector-based routing model (mizuRoute) to generate streamflow at 9579 stream reaches in the sub-continental river basins. The model-simulated streamflow showed good performance against the observed flow with coefficient of determination (R2) and Nash-Sutcliffe efficiency (NSE) above 0.70 for more than 60% of the gauge stations. The dataset was used to examine the variability in low, average, and high flow across the streams. Long-term changes in streamflow showed a significant decline in flow in the Ganga basin while an increase in the semi-arid western India and Indus basin. Long-term streamflow can be used for planning water management and climate change adaptation in the Indian sub-continent.

18. To what extent does river routing matter in hydrological modeling?

Authors: Nicolás Cortés-Salazar; Nicolás Vásquez; Naoki Mizukami; Pablo A. Mendoza; Ximena Vargas

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-27-3505-2023

URL: https://doi.org/10.5194/hess-27-3505-2023

Publication Date: 2023-10-06

Volume/Issue: Volume 27, Issue 19

References: 99

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fondo Nacional de Desarrollo Científico y Tecnológico, Agencia Nacional de Investigación y Desarrollo, National Science Foundation

Open Access: N/A

Abstract: . Spatially distributed hydrology and land surface models are typicallyapplied in combination with river routing schemes that convert instantaneousrunoff into streamflow. Nevertheless, the development of such schemes hasbeen somehow disconnected from hydrologic model calibration research,although both seek to achieve more realistic streamflow simulations. In thispaper, we seek to bridge this gap to understand the extent to which theconfiguration of routing schemes affects hydrologic model parameter searchesin water resources applications. To this end, we configure the Variable Infiltration Capacity (VIC) model coupled with the mizuRoute routing modelin the Cautín River basin (2770 km2), Chile. We use the Latinhypercube sampling (LHS) method to generate 3500 different model parameterssets, for which basin-averaged runoff estimates are obtained directly (norouting or instantaneous runoff case) and are subsequently compared againstoutputs from four routing schemes (unit hydrograph, Lagrangian kinematicwave, Muskingum-Cunge, and diffusive wave) applied with five differentrouting time steps (1, 2, 3, 4, and 6 h). The results show thatincorporating routing schemes may alter streamflow simulations at sub-daily,daily, and even monthly timescales. The maximum Kling-Gupta efficiency (KGE)obtained for daily streamflow increases from 0.64 (instantaneous runoff) to0.81 (for the best routing scheme), and such improvements do not depend onthe routing time step. Moreover, the optimal parameter sets may differdepending on the routing scheme configuration, affecting the baseflowcontribution to total runoff. Including routing models decreases streamflowvalues in flood frequency curves and may alter the probabilisticdistribution of the medium- and low-flow segments of the flow duration curveconsiderably (compared to the case without routing). More generally, theresults presented here highlight the potential impacts of river routingimplementations on water resources applications that involve hydrologicmodels and, in particular, parameter calibration.

19. Real-time forecasting model development work plan

Authors: Francesca Messina; Ioannis Georgiou; Melissa Baustian; Travis Dahl; Jodi Ryder; Michael Miner; Ronald Heath

Journal: Unknown Journal

Publisher: Engineer Research and Development Center (U.S.)

DOI: 10.21079/11681/47599

URL: https://doi.org/10.21079/11681/47599

Publication Date: 2023-09-15

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Time line|Leverage (statistics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The objective of the Lowermost Mississippi River Management Program is to move the nation toward more holistic management of the lower reaches of the Mississippi River through the development and use of a science-based decision-making framework. There has been substantial investment in the last decade to develop multidimensional numerical models to evaluate the Lowermost Mississippi River (LMMR) hydrodynamics, sediment transport, and salinity dynamics. The focus of this work plan is to leverage the existing scientific knowledge and models to improve holistic management of the LMMR. Specifically, this work plan proposes the development of a real-time forecasting (RTF) system for water, sediment, and selected nutrients in the LMMR. The RTF system will help inform and guide the decision-making process for operating flood-control and sediment-diversion structures. This work plan describes the primary components of the RTF system and their interactions. The work plan includes descriptions of the existing tools and numerical models that could be leveraged to develop this system together with a brief inventory of existing real-time data that could be used to validate the RTF system. A description of the tasks that would be required to develop and set up the RTF system is included together with an associated timeline.

20. Unraveling the 2021 Central Tennessee flood event using a hierarchical multi-model inundation modeling framework

Authors: Sudershan Gangrade; Ganesh R. Ghimire; Shih-Chieh Kao; Mario Morales-Hernández; Ahmad A. Tavakoly; Joseph L. Gutenson; Kent H. Sparrow; George K. Darkwah; Alfred J. Kalyanapu; Michael L. Follum

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2023.130157

URL: https://doi.org/10.1016/j.jhydrol.2023.130157

Publication Date: 2023-09-14

Volume/Issue: Volume 625

References: 85

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

21. Machine learning-enabled calibration of river routing model parameters

Authors: Ying Zhao (University of Michigan); Mayank Chadha (University of California San Diego); Nicholas Olsen (c Coastal and Hydraulics Laboratory, US Army Corps of Engineers, Vicksburg, MS, USA); Elissa Yeates (c Coastal and Hydraulics Laboratory, US Army Corps of Engineers, Vicksburg, MS, USA); Josh Turner (d Hottinger Bruel & Kjaer Solutions LLC, Southfield, MI 48076, USA); Guga Gugaratshan (d Hottinger Bruel & Kjaer Solutions LLC, Southfield, MI 48076, USA); Guofeng Qian (University of California San Diego); Michael D. Todd (University of California San Diego); Zhen Hu (University of Michigan)

Journal: Journal of Hydroinformatics

Publisher: IWA Publishing

DOI: 10.2166/hydro.2023.030

URL: https://doi.org/10.2166/hydro.2023.030

Publication Date: 2023-09-01

Volume/Issue: Volume 25, Issue 5

References: 48

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Coastal and Hydraulics Laboratory

Open Access: N/A

Abstract: Streamflow prediction of rivers is crucial for making decisions in watershed and inland waterways management. The US Army Corps of Engineers (USACE) uses a river routing model called RAPID to predict water discharges for thousands of rivers in the network for watershed and inland waterways management. However, the calibration of hydrological streamflow parameters in RAPID is time-consuming and requires streamflow measurement data which may not be available for some ungauged locations. In this study, we aim to address the calibration aspect of the RAPID model by exploring machine learning (ML)-based methods to facilitate efficient calibration of hydrological model parameters without the need for streamflow measurements. Various ML models are constructed and compared to learn a relationship between hydrological model parameters and various river parameters, such as length, slope, catchment size, percentage of vegetation, and elevation contours. The studied ML models include Gaussian process regression, Gaussian mixture copula, Random Forest, and XGBoost. This study has shown that ML models that are carefully constructed by considering causal and sensitive input features offer a potential approach that not only obtains calibrated hydrological model parameters with reasonable accuracy but also bypasses the current calibration challenges.

22. A decadal review of the CREST model family: Developments, applications, and outlook

Authors: Zhi Li; Xianwu Xue; Robert Clark; Humberto Vergara; Jonathan Gourley; Guoqiang Tang; Xinyi Shen; Guangyuan Kan; Ke Zhang; Jiahu Wang; Mengye Chen; Shang Gao; Jiaqi Zhang; Tiantian Yang; Yixin Wen; Pierre Kirstetter; Yang Hong

Journal: Journal of Hydrology X

Publisher: Elsevier BV

DOI: 10.1016/j.hydroa.2023.100159

URL: https://doi.org/10.1016/j.hydroa.2023.100159

Publication Date: 2023-08-23

Volume/Issue: Volume 20

References: 123

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

23. Parallel river channel routing computation based on a straightforward domain decomposition of river networks

Authors: Yong-He Liu; Zong-Liang Yang; Pei-Rong Lin

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2023.129988

URL: https://doi.org/10.1016/j.jhydrol.2023.129988

Publication Date: 2023-07-24

Volume/Issue: Volume 625

References: 32

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

24. Development and evaluation of the channel routing model and parameters within the National Water Model

Authors: L. K. Read (National Center for Atmospheric Research); D. N. Yates (National Center for Atmospheric Research); J. M. McCreight (Cooperative Programs for the Advancement of Earth System Science University Corporation for Atmospheric Research Colorado Boulder USA); A. Rafieeinasab (National Center for Atmospheric Research); K. Sampson (National Center for Atmospheric Research); D. J. Gochis (National Center for Atmospheric Research)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.13134

URL: https://doi.org/10.1111/1752-1688.13134

Publication Date: 2023-06-08

Volume/Issue: Volume 59, Issue 5

References: 45

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Oceanic and Atmospheric Administration

License: N/A

Open Access: N/A

Abstract: The National Water Model (NWM) was deployed by the National Oceanic and Atmospheric Administration to simulate operational forecasts of hydrologic states across the continental United States. This paper describes the geospatial river network ("hydro-fabric"), physics, and parameters of the NWM, elucidating the challenges of extrapolating parameters a large scale with limited observations. A set of regression-based channel geometry parameters are evaluated for a subset of the 2.7 million NWM reaches, and the riverine compound channel scheme is described. Based on the results from regional streamflow experiments within the broader NWM context, the compound channel reduced the root mean squared error by 2% and improved median Nash-Sutcliffe efficiency by 16% compared with a non-compound formulation. Peak event analysis from 910 peak flow events across 26 basins matched from the US Flash Flood Observation Database revealed that the mean timing error is 3 h lagged behind the observations. The routing time step was also tested, for 5-min (default, operational setting) and 1-h increments. The model was computationally stable and able to convey the flood peaks, although the hydrograph shape and peak timing were altered.

25. Construction of a daily streamflow dataset for Peru using a similarity-based regionalization approach and a hybrid hydrological modeling framework

Authors: Harold Llauca; Karen Leon; Waldo Lavado-Casimiro

Journal: Journal of Hydrology: Regional Studies

Publisher: Elsevier BV

DOI: 10.1016/j.ejrh.2023.101381

URL: https://doi.org/10.1016/j.ejrh.2023.101381

Publication Date: 2023-04-12

Volume/Issue: Volume 47

References: 89

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

26. Applicability of a flood forecasting system for Nebraska watersheds

Authors: Sinan Rasiya Koya; Nicolas Velasquez Giron; Marcela Rojas; Ricardo Mantilla; Kirk Harvey; Daniel Ceynar; Felipe Quintero; Witold F. Krajewski; Tirthankar Roy

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2023.105693

URL: https://doi.org/10.1016/j.envsoft.2023.105693

Publication Date: 2023-04-03

Volume/Issue: Volume 164

References: 74

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

27. Higher Frozen Soil Permeability Represented in a Hydrological Model Improves Spring Streamflow Prediction From River Basin to Continental Scales

Authors: Jetal Agnihotri (University of Arizona); Ali Behrangi (University of Arizona); Ahmad Tavakoly (University of Maryland, Coastal and Hydraulics Laboratory US Army Engineer Research and Development Center Vicksburg MS USA); Matthew Geheran (Coastal and Hydraulics Laboratory US Army Engineer Research and Development Center Vicksburg MS USA); Mohammad A. Farmani (University of Arizona); Guo-Yue Niu (University of Arizona)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022wr033075

URL: https://doi.org/10.1029/2022wr033075

Publication Date: 2023-03-26

Volume/Issue: Volume 59, Issue 4

References: 111

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: snow dominated basins, frozen soil, hydrological modeling, streamflow predictions, soil permeability

Funding: NOAA Great Lakes Environmental Research Laboratory, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Despite plentiful evidence of frozen ground effects on snowmelt infiltration from lab experiments at pedon scales, streamflow observations show a weaker or no effect in terms of timing and magnitude at larger scales. We aim to understand this conflicting phenomenon through modeling using the Noah land surface model with multi-physics (MP; Noah-MP) options and the Routing Application for Parallel computatIon of Discharge (RAPID) over the Mississippi River Basin. We conduct 16 experiments with combinations of two supercooled liquid water (SLW) parameterization schemes and four soil hydraulic property schemes in Noah-MP driven by two gridded precipitation products of the North American Land Data Assimilation System (NLDAS) and the Integrated Multi-satellitE Retrievals for GPM (IMERG) Final. We then use RAPID to route Noah-MP modeled surface runoff and groundwater discharge to predict daily streamflow at 52 United States Geological Survey gauges from 2015 to 2019. A model with the highest permeability performs better than other schemes on daily streamflow predictions by 20%-57% throughout a water year and 29%-113% for the spring as measured by the mean Kling-Gupta Efficiency of the 52 gauges. Different SLW schemes demonstrate negligible effects on streamflow predictions. Models forced by IMERG show a better prediction skill compared with those forced by NLDAS at most of the gauges. Both precipitation products confirm that a scheme of higher permeability yields more accurate streamflow predictions over frozen ground. Future models should represent preferential flows through macropore networks to improve the understanding of frozen soil effects on infiltration and discharge across scales.

28. A systematic literature review on lake water level prediction models

Authors: Serkan Ozdemir; Muhammad Yaqub; Sevgi Ozkan Yildirim

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2023.105684

URL: https://doi.org/10.1016/j.envsoft.2023.105684

Publication Date: 2023-03-20

Volume/Issue: Volume 163

References: 110

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation of Korea, Ministry of Science, ICT and Future Planning

License: N/A

Open Access: N/A

Abstract: N/A

29. Possibility of global gridded streamflow dataset correction: applications of large-scale watersheds with different climates

Authors: Hesam Barkhordari; Mohsen Nasseri; Hamidreza Rezazadeh

Journal: Theoretical and Applied Climatology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00704-023-04388-2

URL: https://doi.org/10.1007/s00704-023-04388-2

Publication Date: 2023-02-08

Volume/Issue: Volume 152, Issue 1-2

References: 82

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

30. Future Changes in Climate and Hydroclimate Extremes in East Africa

Authors: S. H. Gebrechorkos (University of Southampton); M. T. Taye (International Water Management Institute (IWMI) Addis Ababa Ethiopia); B. Birhanu (School of Earth Sciences, Addis Ababa University Addis Ababa Ethiopia); D. Solomon (International Livestock Research Institute (ILRI) Addis Ababa Ethiopia); T. Demissie (NORCE Norwegian Research Center Bjerknes Center for Climate Research Bergen Norway, International Livestock Research Institute (ILRI) Addis Ababa Ethiopia)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022ef003011

URL: https://doi.org/10.1029/2022ef003011

Publication Date: 2023-01-30

Volume/Issue: Volume 11, Issue 2

References: 85

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: Climate change is affecting the agriculture, water, and energy sectors in East Africa and the impact is projected to increase in the future. To allow adaptation and mitigation of the impacts, we assessed the changes in climate and their impacts on hydrology and hydrological extremes in East Africa. We used outputs from seven CMIP-6 Global Climate Models (GCMs) and 1981-2010 is used as a reference period. The output from GCMs are statistically downscaled using the Bias Correction-Constructed Analogs with Quantile mapping reordering method to drive a high-resolution hydrological model. The Variable Infiltration Capacity and vector-based routing models are used to simulate runoff and streamflow across 68,300 river reaches in East Africa. The results show an increase in annual precipitation (up to 35%) in Ethiopia, Uganda, and Kenya and a decrease (up to 4.5%) in Southern Tanzania in the 2050s (2041-2070) and 2080s (2071-2100). During the long rainy season (March-May), precipitation is projected to be higher (up to 43%) than the reference period in Southern Ethiopia, Kenya, and Uganda but lower (up to -20%) in Tanzania. Large parts of Kenya, Uganda, Tanzania, and Southern Ethiopia show an increase in precipitation (up to 38%) during the short rainy season (October-December). Temperature and evapotranspiration will continue to increase in the future. Further, annual and seasonal streamflow and hydrological extremes (droughts and floods) are projected to increase in large parts of the region throughout the 21st century calling for site-specific adaptation.

31. Development of non-data driven reservoir routing in the routing application for parallel computatIon of discharge (RAPID) model

Authors: Ahmad A. Tavakoly; Cédric H. David; Joseph L. Gutenson; Mark W. Wahl; Mike Follum

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2023.105631

URL: https://doi.org/10.1016/j.envsoft.2023.105631

Publication Date: 2023-01-28

Volume/Issue: Volume 161

References: 71

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

32. Insights From Dayflow: A Historical Streamflow Reanalysis Dataset for the Conterminous United States

Authors: Ganesh R. Ghimire (Oak Ridge National Laboratory); Carly Hansen (Oak Ridge National Laboratory); Sudershan Gangrade (Oak Ridge National Laboratory); Shih-Chieh Kao (Oak Ridge National Laboratory); Peter E. Thornton (Oak Ridge National Laboratory); Debjani Singh (Oak Ridge National Laboratory)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022wr032312

URL: https://doi.org/10.1029/2022wr032312

Publication Date: 2023-01-17

Volume/Issue: Volume 59, Issue 2

References: 116

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: Dayflow, streamflow reanalysis, GRADES, VIC, RAPID (and 1 more)

Funding: Water Power Technologies Office, U.S. Department of Energy, Office of Science (and 1 more)

Open Access: N/A

Abstract: Reconstructed historical streamflow time series can supplement limited streamflow gauge observations. However, there are common challenges of typical modeling approaches: process-based hydrologic models can be data/computation-intensive, and statistics-based models can be region/stream-specific. Here we present a nationally scalable modeling framework integrating the simulated runoff from the Variable Infiltration Capacity (VIC) model with the Routing Application for Parallel computatIon of Discharge (RAPID) routing model leveraging high-performance computing. We demonstrate an efficient method of assimilating streamflow at US Geological Survey (USGS) streamflow monitoring sites using a simple hierarchical approach in the VIC-RAPID framework. The result is a reconstructed 36-year (1980-2015) daily and monthly streamflow dataset (Dayflow) at ~2.7 million NHDPlusV2 stream reaches in the conterminous US (CONUS). We perform a comprehensive evaluation at 7,526 USGS sites and characterize their error statistics. The results demonstrate that 49% of the USGS sites demonstrate Kling-Gupta Efficiency (KGE) > 0.5 and 58% of the sites show percentage bias within +/-20% for the daily naturalized streamflow. Streamflow data assimilation across CONUS shows an overall improvement over naturalized streamflow, notably in the western semiarid-to-arid regions. Comparison to other national and global streamflow reanalysis datasets such as the National Water Model and Global Reach-scale A priori Discharge Estimates for SWOT demonstrates improved KGE, reduced bias, and directions for Dayflow improvements. Investigations of error statistics with key hydrologic, hydroclimatic, and geomorphologic basin characteristics reveal region-specific patterns which may help improve future framework applications. Overall, Dayflow may enable a better understanding of hydrologic conditions in a changing environment, especially in locations currently not represented by streamflow monitoring networks.

33. Cooperative adaptive management of the Nile River with climate and socio-economic uncertainties

Authors: Mohammed Basheer; Victor Nechifor; Alvaro Calzadilla; Solomon Gebrechorkos; David Pritchard; Nathan Forsythe; Jose M. Gonzalez; Justin Sheffield; Hayley J. Fowler; Julien J. Harou

Journal: Nature Climate Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41558-022-01556-6

URL: https://doi.org/10.1038/s41558-022-01556-6

Publication Date: 2023-01-09

Volume/Issue: Volume 13, Issue 1

References: 88

Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: RCUK | Economic and Social Research Council

Open Access: N/A

Abstract: The uncertainties around the hydrological and socio-economic implications of climate change pose a challenge for Nile River system management, especially with rapidly rising demands for river-system-related services and political tensions between the riparian countries. Cooperative adaptive management of the Nile can help alleviate some of these stressors and tensions. Here we present a planning framework for adaptive management of the Nile infrastructure system, combining climate projections; hydrological, river system and economy-wide simulators; and artificial intelligence multi-objective design and machine learning algorithms. We demonstrate the utility of the framework by designing a cooperative adaptive management policy for the Grand Ethiopian Renaissance Dam that balances the transboundary economic and biophysical interests of Ethiopia, Sudan and Egypt. This shows that if the three countries compromise cooperatively and adaptively in managing the dam, the national-level economic and resilience benefits are substantial, especially under climate projections with the most extreme streamflow changes.

34. Improvement of low flows simulation in the SASER hydrological modeling chain

Authors: Omar Cenobio-Cruz; Pere Quintana-Seguí; Anaïs Barella-Ortiz; Ane Zabaleta; Luis Garrote; Roger Clavera-Gispert; Florence Habets; Santiago Beguería

Journal: Journal of Hydrology X

Publisher: Elsevier BV

DOI: 10.1016/j.hydroa.2022.100147

URL: https://doi.org/10.1016/j.hydroa.2022.100147

Publication Date: 2022-12-15

Volume/Issue: Volume 18

References: 84

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Faculty of Science and Engineering, University of Manchester, Federación Española de Enfermedades Raras

License: N/A

Open Access: N/A

Abstract: N/A

35. Climate change projections of continental-scale streamflow across the Mississippi River Basin

Authors: James W. Lewis; Sara E. Lytle; Ahmad A. Tavakoly

Journal: Theoretical and Applied Climatology

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00704-022-04243-w

URL: https://doi.org/10.1007/s00704-022-04243-w

Publication Date: 2022-12-13

Volume/Issue: Volume 151, Issue 3-4

References: 49

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Army Corps of Engineers

Open Access: N/A

Abstract: A large body of scientific research has demonstrated a changing climate, which affects river flow regimes and extreme flood frequencies and magnitudes. The magnitude and frequency of extreme events are of critical importance in the evaluation of river systems to inform flood risk reduction under current and future conditions. The global climate projections from the Coupled Model Intercomparison Project, Phase 5 (CMIP5) datasets were used by the Variable Infiltration Capacity (VIC) land surface model to produce a runoff dataset, implementing a Bias-Correction Spatial Disaggregation (BCSD) approach. The resulting runoff was then used as input to the Routing Application for Parallel computatIon of Discharge (RAPID) river routing model to simulate daily flows within all 1.2 million Mississippi River Basin river reaches for years 1950 through 2099. This research effort analyzed the performance of the models for the historical time period, comparing with the observations at 64 gage locations for 16 different climate models. A recurrence interval analysis was performed to determine the 2-, 5-, 10-, 50-, 100-, 500-, and 1000-year events within both the historical and projected time periods, highlighting the relative changes predicted into the future. Anticipated seasonal changes are demonstrated by comparing monthly average streamflows for three different time periods (1951-2005, 2006-2049, and 2050-2099). Results indicate that the hydrologic conditions of the Lower Mississippi River are not stationary. Based on all 16 models considered in this study, the median of the model projections shows an 8% increase in the 100-year return period discharge at Vicksburg, Mississippi, into the future time period, although the full range of 16 models varies widely from - 11 to + 85% change in the 100-year discharge in the future.

36. USUAL Watershed Tools: A new geospatial toolkit for hydro-geomorphic delineation

Authors: Scott R. David; Brendan P. Murphy; Jonathan A. Czuba; Muneer Ahammad; Patrick Belmont

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2022.105576

URL: https://doi.org/10.1016/j.envsoft.2022.105576

Publication Date: 2022-11-05

Volume/Issue: Volume 159

References: 47

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Joint Fire Science Program, National Science Foundation, USDA (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

37. Intercomparison of global ERA reanalysis products for streamflow simulations at the high-resolution continental scale

Authors: R.L. Bain; M.J. Shaw; M.P. Geheran; A.A. Tavakoly; M.D. Wahl; E. Zsoter

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2022.128624

URL: https://doi.org/10.1016/j.jhydrol.2022.128624

Publication Date: 2022-10-28

Volume/Issue: Volume 616

References: 75

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

38. A graph neural network (GNN) approach to basin-scale river network learning: the role of physics-based connectivity and data fusion

Authors: Alexander Y. Sun; Peishi Jiang; Zong-Liang Yang; Yangxinyu Xie; Xingyuan Chen

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-26-5163-2022

URL: https://doi.org/10.5194/hess-26-5163-2022

Publication Date: 2022-10-14

Volume/Issue: Volume 26, Issue 19

References: 116

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Biological and Environmental Research, Advanced Scientific Computing Research

Open Access: N/A

Abstract: . Rivers and river habitats around the world are under sustained pressurefrom human activities and the changing global environment. Our abilityto quantify and manage the river states in a timely manner is criticalfor protecting the public safety and natural resources. In recentyears, vector-based river network models have enabled modeling oflarge river basins at increasingly fine resolutions, but are computationally demanding. This work presents a multistage, physics-guided, graph neural network (GNN) approach for basin-scale river network learningand streamflow forecasting. During training, we train a GNN modelto approximate outputs of a high-resolution vector-based river networkmodel; we then fine-tune the pretrained GNN model with streamflowobservations. We further apply a graph-based, data-fusion step tocorrect prediction biases. The GNN-based framework is first demonstratedover a snow-dominated watershed in the western United States. A series of experiments are performed to test different training and imputation strategies. Results show that the trained GNN model can effectively serve as a surrogate of the process-based model with high accuracy, with medianKling-Gupta efficiency (KGE) greater than 0.97. Application of thegraph-based data fusion further reduces mismatch between the GNN modeland observations, with as much as 50 % KGE improvement oversome cross-validation gages. To improve scalability, a graph-coarseningprocedure is introduced and is demonstrated over a much larger basin.Results show that graph coarsening achieves comparable prediction skillsat only a fraction of training cost, thus providing important insightsinto the degree of physical realism needed for developing large-scaleGNN-based river network models.

39. Advancing global hydrologic modeling with the GEOGloWS ECMWF streamflow service

Authors: Riley C. Hales (Brigham Young University); E. James Nelson (Brigham Young University); Michael Souffront (Aquaveo LLC Provo Utah USA); Angelica L. Gutierrez (Office of Water Prediction National Oceanic and Atmospheric Administration Washington District of Columbia USA); Christel Prudhomme (European Centre for Medium Range Weather Forecasting Reading UK); Steve Kopp (Esri Redlands California USA); Daniel P. Ames (Brigham Young University); Gustavious P. Williams (Brigham Young University); Norman L. Jones (Brigham Young University)

Journal: Journal of Flood Risk Management

Publisher: Wiley

DOI: 10.1111/jfr3.12859

URL: https://doi.org/10.1111/jfr3.12859

Publication Date: 2022-10-06

Volume/Issue: Volume 18, Issue 1

References: 87

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Microsoft, World Bank Group

Open Access: N/A

Abstract: Most people face some level of water insecurity. Wise water management practices to address water security issues typically require data derived from a combination of observation and model data. This data has historically proven difficult to sustainably supply in many areas of the world. We present the design and development of a global, modeled streamflow data source for the Group on Earth Observation (GEO) Global Water Sustainability (GEOGloWS) implemented at the European Centre for Medium-Range Weather Forecasts (ECMWF). This GEOGloWS ECMWF Streamflow Service (GEOGloWS Service) is a solution and prototype to sustainably address this need for data. The GEOGloWS Service centralizes computing and human resources to build a global hydrologic model and exposes data and mapping web services that allow users to consume the resulting data to meet their specific needs. The global hydrologic model produces global 15-day ensemble streamflow forecasts and a historical simulation since January 1979. We present case studies in several countries and research environments which demonstrate the utility of the approach taken by the GEOGloWS Service. The case studies show how the global modeled data are being applied to make informed decisions and advance projects in ways that otherwise would not have been possible.

40. CREST-VEC: a framework towards more accurate and realistic flood simulation across scales

Authors: Zhi Li; Shang Gao; Mengye Chen; Jonathan Gourley; Naoki Mizukami; Yang Hong

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-6181-2022

URL: https://doi.org/10.5194/gmd-15-6181-2022

Publication Date: 2022-08-08

Volume/Issue: Volume 15, Issue 15

References: 76

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . Large-scale (i.e., continental and global) hydrologicsimulation is an appealing yet challenging topic for the hydrologiccommunity. First and foremost, model efficiency and scalability (flexibilityin resolution and discretization) have to be prioritized. Then, sufficientmodel accuracy and precision are required to provide useful information forwater resource applications. Towards this goal, we craft two objectives forimproving US current operational hydrological models: (1) vectorized routingand (2) improved hydrological processes. This study presents a hydrologicmodeling framework, CREST-VEC, that combines a gridded water balance model anda newly developed vector-based routing scheme. First, in contrast to aconventional fully gridded model, this framework can significantly reducethe computational cost of river routing by at least 10 times, based onexperiments at regional (0.07 vs. 0.002 s per step) and continentalscales (0.35 vs. 7.2 s per step). This provides adequate timeefficiency for generating operational ensemble streamflow forecasts and evenprobabilistic estimates across scales. Second, the performance using the newvector-based routing is improved, with the median-aggregated NSE(Nash-Sutcliffe efficiency) score increasing from -0.06 to 0.18 over theCONUS (contiguous US). Third, with the lake module incorporated, the NSE score is furtherimproved by 56.2 % and the systematic bias is reduced by 17 %. Lastly,over 20 % of the false alarms on 2-year floods in the US can bemitigated with the lake module enabled, at the expense of only missing2.3 % more events. This study demonstrated the advantages of the proposedhydrological modeling framework, which could provide a solid basis forcontinental- and global-scale water modeling at fine resolution. Furthermore,the use of ensemble forecasts can be incorporated into this framework; andthus, optimized streamflow prediction with quantified uncertaintyinformation can be achieved in an operational fashion for stakeholders anddecision-makers.

41. The impact of multi-sensor land data assimilation on river discharge estimation

Authors: Wen-Ying Wu; Zong-Liang Yang; Long Zhao; Peirong Lin

Journal: Remote Sensing of Environment

Publisher: Elsevier BV

DOI: 10.1016/j.rse.2022.113138

URL: https://doi.org/10.1016/j.rse.2022.113138

Publication Date: 2022-06-30

Volume/Issue: Volume 279

References: 67

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

42. Hydrokinetic energy conversion: A global riverine perspective

Authors: Michael Ridgill (School of Ocean Sciences, Bangor University 1 , Menai Bridge LL59 5AB, United Kingdom); Matt J. Lewis (School of Ocean Sciences, Bangor University 1 , Menai Bridge LL59 5AB, United Kingdom); Peter E. Robins (School of Ocean Sciences, Bangor University 1 , Menai Bridge LL59 5AB, United Kingdom); Sopan D. Patil (School of Natural Sciences, Bangor University 2 , Bangor LL57 2UW, United Kingdom); Simon P. Neill (School of Ocean Sciences, Bangor University 1 , Menai Bridge LL59 5AB, United Kingdom)

Journal: Journal of Renewable and Sustainable Energy

Publisher: AIP Publishing

DOI: 10.1063/5.0092215

URL: https://doi.org/10.1063/5.0092215

Publication Date: 2022-06-29

Volume/Issue: Volume 14, Issue 4

References: 77

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Social Fund, European Regional Development Fund

License: N/A

Open Access: N/A

Abstract: Free-flowing rivers have been impacted by anthropogenic activity and extensive hydropower development. Despite this, many opportunities exist for context-specific energy extraction, at locations deemed undesirable for conventional hydropower plants, in ways that reduce the scale of operation and impact. Hydrokinetic energy conversion is a renewable energy technology that requires accurate resource assessment to support deployment in rivers. We use global-scale modeled river discharge data, combined with a high-resolution vectorized representation of river networks, to estimate channel form, flow velocities, and, hence, global hydrokinetic potential. Our approach is based directly on the transfer of kinetic energy through the river network, rather than conventional, yet less realistic, assessments that are based on conversion from gravitational potential energy. We show that this new approach provides a more accurate global distribution of the hydrokinetic resource, highlighting the importance of the lower-courses of major rivers. The resource is shown to have great potential on the continents of South America, Asia, and Africa. We calculate that the mean hydrokinetic energy of global rivers (excluding Greenland and Antarctica) is 5.911 +/- 0.009 PJ (1.642 +/- 0.003 TWh).

43. Variability and changes in hydrological drought in the Volta Basin, West Africa

Authors: Solomon H. Gebrechorkos; Ming Pan; Peirong Lin; Daniela Anghileri; Nathan Forsythe; David M.W. Pritchard; Hayley J. Fowler; Emmanuel Obuobie; Deborah Darko; Justin Sheffield

Journal: Journal of Hydrology: Regional Studies

Publisher: Elsevier BV

DOI: 10.1016/j.ejrh.2022.101143

URL: https://doi.org/10.1016/j.ejrh.2022.101143

Publication Date: 2022-06-17

Volume/Issue: Volume 42

References: 101

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UK Research and Innovation

License: N/A

Open Access: N/A

Abstract: N/A

44. Impact of Riverine Fresh Water on Indian Summer Monsoon: Coupling a Runoff Routing Model to a Global Seasonal Forecast Model

Authors: Ankur Srivastava; Suryachandra A. Rao; Subimal Ghosh

Journal: Frontiers in Climate

Publisher: Frontiers Media SA

DOI: 10.3389/fclim.2022.902586

URL: https://doi.org/10.3389/fclim.2022.902586

Publication Date: 2022-06-14

Volume/Issue: Volume 4

References: 123

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: Rivers form an essential component of the earth system, with ~36,000 km3 of riverine freshwater being dumped into the global oceans every year. The role of rivers in controlling the sea-surface salinity and ensuing air-sea interactions in the Bay of Bengal (BoB) is well-known from observational studies; however, attempts to include rivers in coupled models used for seasonal prediction have been limited. This study reports the benefits of river routing in coupled models over prescribing observational river discharge and the impact on the Indian Summer Monsoon (ISM) simulation. Seasonal hindcasts are carried out using a state-of-the-art global coupled ocean-atmosphere-land-sea ice model, Climate Forecast System version 2, coupled to a runoff routing model. It is demonstrated that such a coupling leads to a better representation of the upper ocean stratification in northern BoB, causes mixed layer warming during July-August, and imparts a significant inter-annual variability to the mixed layer heat budget. The rainfall-runoff coupled feedback associated with ISM is captured better, and remote teleconnections with the equatorial Pacific are enhanced. Improved seasonal mean temperature and salinity profiles in the northern BoB lead to the formation of a thicker barrier layer, which is closely tied to the freshwater from rivers. These processes result in an overall enhancement of the ISM rainfall simulation skill, which stems from scale interactions between the sub-seasonal and seasonal variability of ISM. A significant community effort is required to reduce biases in land-surface processes to improve streamflow simulations, along with better parameterization of mixing of river water with the ocean.

45. River network and hydro-geomorphological parameters at 1■12 degrees resolution for global hydrological and climate studies

Authors: Simon Munier; Bertrand Decharme

Journal: Earth System Science Data

Publisher: Copernicus GmbH

DOI: 10.5194/essd-14-2239-2022

URL: https://doi.org/10.5194/essd-14-2239-2022

Publication Date: 2022-05-12

Volume/Issue: Volume 14, Issue 5

References: 73

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . Global-scale river routing models (RRMs) are commonly used in a variety of studies, including studies on the impact of climate change on extreme flows (floods and droughts), water resources monitoring or large-scale flood forecasting. Over the last two decades, the increasing number of observational datasets, mainly from satellite missions, and increasing computing capacities have allowed better performance by RRMs, namely by increasing their spatial resolution. The spatial resolution of a RRM corresponds to the spatial resolution of its river network, which provides the flow directions of all grid cells. River networks may be derived at various spatial resolutions by upscaling high-resolution hydrography data.This paper presents a new global-scale river network at 1/12■ derived from the MERIT-Hydro dataset. The river network is generated automatically using an adaptation of the hierarchical dominant river tracing (DRT) algorithm, and its quality is assessed over the 70 largest basins of the world. Although this new river network may be used for a variety of hydrology-related studies, it is provided here with a set of hydro-geomorphological parameters at the same spatial resolution. These parameters are derived during the generation of the river network and are based on the same high-resolution dataset, so that the consistency between the river network and the parameters is ensured. The set of parameters includes a description of river stretches (length, slope, width, roughness, bankfull depth), floodplains (roughness, sub-grid topography) and aquifers (transmissivity, porosity, sub-grid topography).The new river network and parameters are assessed by comparing the performances of two global-scale simulations with the CTRIP model, one with the current spatial resolution (1/2■) and the other with the new spatial resolution (1/12■). It is shown that, overall, CTRIP at 1/12■ outperforms CTRIP at 1/2■, demonstrating the added value of the spatial resolution increase.The new river network and the consistent hydro-geomorphology parameters, freely available for download from Zenodo (https://doi.org/10.5281/zenodo.6482906, Munier and Decharme, 2022), may be useful for the scientific community, especially for hydrology and hydro-geology modelling, water resources monitoring or climate studies.

46. An integrated approach to streamflow estimation and flood inundation mapping using VIC, RAPID and LISFLOOD-FP

Authors: Saswata Nandi; Manne Janga Reddy

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2022.127842

URL: https://doi.org/10.1016/j.jhydrol.2022.127842

Publication Date: 2022-04-21

Volume/Issue: Volume 610

References: 61

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

47. HydroBlocks v0.2: enabling a field-scale two-way coupling between the land surface and river networks in Earth system models

Authors: Nathaniel W. Chaney; Laura Torres-Rojas; Noemi Vergopolan; Colby K. Fisher

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-6813-2021

URL: https://doi.org/10.5194/gmd-14-6813-2021

Publication Date: 2021-11-09

Volume/Issue: Volume 14, Issue 11

References: 53

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Climate Program Office

Open Access: N/A

Abstract: . Over the past decade, there has been appreciable progresstowards modeling the water, energy, and carbon cycles at field scales(10-100 m) over continental to global extents in Earth system models(ESMs). One such approach, named HydroBlocks, accomplishes this task whilemaintaining computational efficiency via Hydrologic Response Units (HRUs),more commonly known as "tiles" in ESMs. In HydroBlocks, these HRUs arelearned via a hierarchical clustering approach from available globalhigh-resolution environmental data. However, until now there has yet to bea river routing approach that is able to leverage HydroBlocks' approach tomodeling field-scale heterogeneity; bridging this gap will make it possibleto more formally include riparian zone dynamics, irrigation from surfacewater, and interactive floodplains in the model. This paper introduces anovel dynamic river routing scheme in HydroBlocks that is intertwined withthe modeled field-scale land surface heterogeneity. Each macroscale polygon(a generalization of the concept of macroscale grid cell) is assigned itsown fine-scale river network that is derived from very high resolution(~ 30 m) digital elevation models (DEMs); the inlet-outlet reaches of a domain'smacroscale polygons are then linked to assemble a full domain's rivernetwork. The river dynamics are solved at the reach-level via the kinematicwave assumption of the Saint-Venant equations. Finally, a two-way couplingbetween each HRU and its corresponding fine-scale river reaches isestablished. To implement and test the novel approach, a 1.0■ boundingbox surrounding the Atmospheric Radiation and Measurement (ARM) SouthernGreat Plains (SGP) site in northern Oklahoma (United States) is used. Theresults show (1) the implementation of the two-way coupling between the landsurface and the river network leads to appreciable differences in thesimulated spatial heterogeneity of the surface energy balance, (2) a limitednumber of HRUs (~ 300 per 0.25■ cell) are required toapproximate the fully distributed simulation adequately, and (3) the surfaceenergy balance partitioning is sensitive to the river routing modelparameters. The resulting routing scheme provides an effective and efficientpath forward to enable a two-way coupling between the high-resolution rivernetworks and state-of-the-art tiling schemes in ESMs.

48. Efficient Metamodel Approach to Handling Constraints in Nonlinear Optimization for Drought Management

Authors: Tingting Zhao (Applied Scientist, Microsoft Corporate, One Microsoft Way, Redmond, WA 98052 (corresponding author).); Barbara Minsker (Professor and Department Chair, Dept. of Civil and Environmental Engineering, Southern Methodist Univ., Dallas, TX 75275.)

Journal: Journal of Water Resources Planning and Management

Publisher: American Society of Civil Engineers (ASCE)

DOI: 10.1061/(asce)wr.1943-5452.0001476

URL: https://doi.org/10.1061/(asce)wr.1943-5452.0001476

Publication Date: 2021-09-27

Volume/Issue: Volume 147, Issue 12

References: 45

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

49. Service-Driven Modeling Approach to Managing Water Allocation in Priority Doctrine Regions

Authors: Tingting Zhao (Applied Scientist, Microsoft Corporate, One Microsoft Way, Redmond, WA 98052 (corresponding author).); Barbara Minsker (Professor and Department Chair, Dept. of Civil and Environmental Engineering, Southern Methodist Univ., Dallas, TX 75275.); Jacob Spoelstra (Director of Data Science, Microsoft Corporate, One Microsoft Way, Redmond, WA 98052.); Christopher Navarro (Lead Research Programmer, National Center for Super Computing Applications, 1205 W. Clark St., Urbana, IL 61801.); Jong Lee (Deputy Associate Director, National Center for Super Computing Applications, 1205 W. Clark St., Urbana, IL 61801.)

Journal: Journal of Water Resources Planning and Management

Publisher: American Society of Civil Engineers (ASCE)

DOI: 10.1061/(asce)wr.1943-5452.0001463

URL: https://doi.org/10.1061/(asce)wr.1943-5452.0001463

Publication Date: 2021-09-02

Volume/Issue: Volume 147, Issue 11

References: 53

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

50. Verification of two hydrological models for real-time flood forecasting in the Hindu Kush Himalaya (HKH) region

Authors: Karma Tsering; Manish Shrestha; Kiran Shakya; Birendra Bajracharya; Mir Matin; Jorge Luis Sanchez Lozano; Jim Nelson; Tandin Wangchuk; Binod Parajuli; Md Arifuzzaman Bhuyan

Journal: Natural Hazards

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11069-021-05014-y

URL: https://doi.org/10.1007/s11069-021-05014-y

Publication Date: 2021-09-01

Volume/Issue: Volume 110, Issue 3

References: 28

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: USAID Bureau for Resilience and Food Security

Open Access: N/A

Abstract: The Hindu Kush Himalayan region is extremely susceptible to periodic monsoon floods. Early warning systems with the ability to predict floods in advance can benefit tens of millions of people living in the region. Two web-based flood forecasting tools (ECMWF-SPT and HIWAT-SPT) are therefore developed and deployed jointly by SERVIR-HKH and NASA-AST to provide early warning to Bangladesh, Bhutan, and Nepal. ECMWF-SPT provides ensemble forecast up to 15-day lead time, whereas HIWAT-SPT provides deterministic forecast up to 3-day lead time covering almost 100% of the rivers. Hydrological models in conjunction with forecast validation contribute not only to advancing the processes of a forecasting system, but also objectively assess the joint distribution of forecasts and observations in quantifying forecast accuracy. The validation of forecast products has emerged as a priority need to evaluate the worth of the predictive information in terms of quality and consistency. This paper describes the effort made in developing the hydrological forecast systems, the current state of the flood forecast services, and the performance of the forecast evaluation. Both tools are validated using a selection of appropriate metrics in measurement in both probabilistic and deterministic space. The numerical metrics are further complemented by graphical representations of scores and probabilities. It was found that the models had a good performance in capturing high flood events. The evaluation across multiple locations indicates that the model performance and forecast goodness are variable on spatiotemporal scale. The resulting information is used to support good decision-making in risk and resource management.

51. Direct Integration of Numerous Dams and Reservoirs Outflow in Continental Scale Hydrologic Modeling

Authors: Ahmad A. Tavakoly (University of Maryland, US Army Engineer Research and Development Center Coastal and Hydraulics Laboratory Vicksburg MS USA); Joseph L. Gutenson (University of Alabama, US Army Engineer Research and Development Center Coastal and Hydraulics Laboratory Vicksburg MS USA); James W. Lewis (U.S. Army Corps of Engineers Mississippi Valley Division Vicksburg MS USA); Michael L. Follum (Wyoming Area Office US Bureau of Reclamation Mills WY USA); Adnan Rajib (Department of Environmental Engineering Texas A&M; University Kingsville TX USA); William Clay LaHatte (US Army Engineer Research and Development Center Coastal and Hydraulics Laboratory Vicksburg MS USA); Chase O. Hamilton (US Army Engineer Research and Development Center Coastal and Hydraulics Laboratory Vicksburg MS USA)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020wr029544

URL: https://doi.org/10.1029/2020wr029544

Publication Date: 2021-08-31

Volume/Issue: Volume 57, Issue 9

References: 73

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: continental scale hydrological modeling, rivers/streams, flood inundation mapping, reservoir impacts, hyper resolution

Funding: Engineer Research and Development Center

License: N/A

Open Access: N/A

Abstract: Despite the recent developments in continental-scale streamflow and flood inundation modeling frameworks, effects of time-specific and spatially explicit storage-release dynamics of numerous dams and reservoirs remain underexplored. This paper fills this knowledge gap by directly inserting operational daily flow release data at 175 dam locations into a streamflow simulation of ~1.2 million river reaches in the Mississippi River Basin (MRB), and therefore quantifying the effect of these regulations on streamflow and flood inundation extents. Using a streamflow routing model called the Routing Application for Parallel computatIon of Discharge (RAPID) and flood inundation mapping model called AutoRoute, two simulation scenarios were constructed respectively including and excluding the daily flow releases from those dams and reservoirs for a 10-year period (2005-2014). Flood inundation maps were simulated for peak flow conditions at a ~10-m hyper spatial resolution. Kling-Gupta efficiency (KGE) values show that streamflow model performance considerably improved when reservoirs were included in the modeled system, varying from 2% in the eastern region to 380% in the drier western region. Despite small variation of streamflow model improvement with reservoir release inclusion in the eastern region of the basin, the flow model was able to better capture observed peak flows. For a 1% change in streamflow, we observe a 0.8% change in estimated flood inundation. Comparisons to three observed flood events in the MRB demonstrate that the flood inundation estimates improve when percent change in streamflow is relatively high. Overall, inclusion of reservoir

release resulted in substantial improvement in continental-scale streamflow and flood inundation mapping.

52. Hydrostreamer v1.0 - improved streamflow predictions for local applications from an ensemble of downscaled global runoff products

Authors: Marko Kallio; Joseph H. A. Guillaume; Vili Virkki; Matti Kummu; Kirsi Virrantaus

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-5155-2021

URL: https://doi.org/10.5194/gmd-14-5155-2021

Publication Date: 2021-08-18

Volume/Issue: Volume 14, Issue 8

References: 83

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Research Council of Finland, Emil Aaltosen Säätiö, H2020 European Research Council (and 1 more)

Open Access: N/A

Abstract: . An increasing number of different types of hydrological,land surface, and rainfall-runoff models exist to estimate streamflow inriver networks. Results from various model runs from global to local scalesare readily available online. However, the usability of these products isoften limited, as they often come aggregated in spatial units which are notcompatible with the desired analysis purpose. We present here an R package,a software library Hydrostreamer v1.0, which aims to improve the usability of existing runoffproducts by addressing the modifiable area unit problem and allowsnon-experts with little knowledge of hydrology-specific modelling issues andmethods to use them for their analyses. Hydrostreamer workflow includes (1) interpolation from source zones to target zones, (2) river routing, and (3) data assimilation via model averaging, given multiple input runoff andobservation data. The software implements advanced areal interpolationmethods and area-to-line interpolation not available in other products andis the first R package to provide vector-based routing. Hydrostreamer iskept as simple as possible - intuitive with minimal data requirements -and minimises the need for calibration. We tested the performance ofHydrostreamer by downscaling freely available coarse-resolution globalrunoff products from the Inter-Sectoral Impact Model Intercomparison Project(ISIMIP) in an application in 3S Basin in Southeast Asia. Results arecompared to observed discharges as well as two benchmark streamflow dataproducts, finding comparable or improved performance. Hydrostreamer v1.0 isopen source and is available from http://github.com/mkkallio/hydrostreamer/ (last access: 5 May 2021) under the MIT licence.

53. Modeling Daily Floods in the Lancang-Mekong River Basin Using an Improved Hydrological-Hydrodynamic Model

Authors: Jie Wang (Chinese Academy of Sciences); Xiaobo Yun (Chinese Academy of Sciences); Yadu Pokhrel (Michigan State University); Dai Yamazaki (University of Tokyo); Qiudong Zhao (State Key Laboratory of Cryospheric Science Northwest Institute of Eco-Environment and Resources Chinese Academic of Sciences Lanzhou China, Key Laboratory of Ecohydrology of Inland River Basin Northwest Institute of Eco-Environment and Resources Chinese Academy of Science Lanzhou China); Aifang Chen (Southern University of Science and Technology); Qiuhong Tang (Chinese Academy of Sciences)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021wr029734

URL: https://doi.org/10.1029/2021wr029734

Publication Date: 2021-07-26

Volume/Issue: Volume 57, Issue 8

References: 58

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: Lancang-Mekong river basin, reservoir regulation, regional parameterization, daily floods, hydrological-hydrodynamic model

Funding: National Natural Science Foundation of China, National Science Foundation, China Postdoctoral Science Foundation

License: N/A

Open Access: N/A

Abstract: Daily floods including event, characteristic, extreme and inundation in the Lancang-Mekong River Basin (LMRB), crucial for flood projection and forecasting, have not been adequately modeled. An improved hydrological-hydrodynamic model (VIC and CaMa-Flood) considering regional parameterization was developed to simulate the flood dynamics over the basin from 1967 to 2015. The flood elements were extracted from daily time series and evaluated at both local and regional scales using the data collected from in-situ observations and remote sensing. The results show that the daily discharge and water level are both well simulated at selected stations with relative error (RE) less than 10% and Nash-Sutcliffe efficiency coefficient (NSE) over 0.90. Half of the flood events have NSE exceeding 0.76. The peak time and flood volume are well reproduced while both peak discharge and water level are slightly underestimated. The results tend to worsen when the characteristics of flood events are extended to annual extremes. These extremes are generally underestimated with NSE less than 0.5 but RE is within 20%. The simulated rainy season inundation area generally agrees with observations from remote sensing, with about 86.8% inundation occurrence frequency captured within the model capacity. Ignoring the regional parameterization and reservoir regulation can both deteriorate flood simulation performance at the local scale, resulting in lower NSE. Specifically, systematically higher water levels and up to 27% overestimation of peak discharge are found when ignoring regional parameterization, while ignoring reservoir regulation would cause up to 23% overestimation for flood extremes. It is expected that these findings would contribute to the regional flood forecasting and flood

management.

54. Global Reach-Level 3-Hourly River Flood Reanalysis (1980-2019)

Authors: Yuan Yang (Princeton University, Tsinghua University); Ming Pan (Scripps Institution of Oceanography / University of California San Diego, Princeton University); Peirong Lin (Peking University); Hylke E. Beck (European Commission); Zhenzhong Zeng (Southern University of Science and Technology); Dai Yamazaki (University of Tokyo); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hui Lu (Tsinghua University); Kun Yang (Tsinghua University); Yang Hong (University of Oklahoma); Eric F. Wood (Princeton University)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

DOI: 10.1175/bams-d-20-0057.1

URL: https://doi.org/10.1175/bams-d-20-0057.1

Publication Date: 2021-07-08

Volume/Issue: Volume 102, Issue 11

References: 100

Cited by: 47 publications

Language: N/A

Subjects: N/A

Keywords: Hydrologic models, Risk assessment, Rivers, Flood events, Hydrology

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Better understanding and quantification of river floods for very local and "flashy" events calls for modeling capability at fine spatial and temporal scales. However, long-term discharge records with a global coverage suitable for extreme events analysis are still lacking. Here, grounded on recent breakthroughs in global runoff hydrology, river modeling, high-resolution hydrography, and climate reanalysis, we developed a 3-hourly river discharge record globally for 2.94 million river reaches during the 40-yr period of 1980-2019. The underlying modeling chain consists of the VIC land surface model (0.05 degrees, 3-hourly) that is well calibrated and bias corrected and the RAPID routing model (2.94 million river and catchment vectors), with precipitation input from MSWEP and other meteorological fields downscaled from ERA5. Flood events (above 2-yr return) and their characteristics (number, spatial distribution, and seasonality) were extracted and studied. Validations against 3-hourly flow records from 6,000+ gauges in CONUS and daily records from 14,000+ gauges globally show good modeling performance across all flow ranges, good skills in reconstructing flood events (high extremes), and the benefit of (and need for) subdaily modeling. This data record, referred as Global Reach-Level Flood Reanalysis (GRFR), is publicly available at https://www.reachhydro.org/home/records/grfr.

55. A container-based approach for sharing environmental models as web services

Authors: Xiaohui Qiao (Brigham Young University); Zhiyu Li (Brigham Young University); Fengyuan Zhang (Key Laboratory of Virtual Geographic Environment (Ministry of Education of PRC), Nanjing Normal University, Nanjing, People's Republic of China, State Key Laboratory Cultivation Base of Geographical Environment Evolution (Jiangsu Province), Nanjing, People's Republic of China, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application, Nanjing, People's Republic of China); Daniel P. Ames (Brigham Young University); Min Chen (Key Laboratory of Virtual Geographic Environment (Ministry of Education of PRC), Nanjing Normal University, Nanjing, People's Republic of China, State Key Laboratory Cultivation Base of Geographical Environment Evolution (Jiangsu Province), Nanjing, People's Republic of China, Jiangsu Center for Collaborative Innovation in Geographical Information Resource Development and Application, Nanjing, People's Republic of China); E. James Nelson (Brigham Young University); Rohit Khattar (Brigham Young University)

Journal: International Journal of Digital Earth

Publisher: Informa UK Limited

DOI: 10.1080/17538947.2021.1925758

URL: https://doi.org/10.1080/17538947.2021.1925758

Publication Date: 2021-05-26

Volume/Issue: Volume 14, Issue 8

References: 54

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Fund for Excellent Young Scholars of China, NASA SERVIR

License: N/A

Open Access: N/A

Abstract: N/A

56. Quantifying the potential of AQPI gap-filling radar network for streamflow simulation through a WRF-Hydro experiment

Authors: Yingzhao Ma (Colorado State University); V. Chandrasekar (Colorado State University); Haonan Chen (Colorado State University); Robert Cifelli (2 NOAA/Physical Sciences Laboratory, Boulder, CO 80305, USA)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm-d-20-0122.1

URL: https://doi.org/10.1175/jhm-d-20-0122.1

Publication Date: 2021-05-18

Volume/Issue: N/A

References: N/A

Cited by: 3 publications

Language: N/A

Subjects: N/A

Keywords: North America, Hydrometeorology, Radars/Radar observations, Hydrologic models

Funding: N/A

License: N/A

Open Access: N/A

Abstract: It remains a challenge to provide accurate and timely flood warnings in many parts of the western United States. As part of the Advanced Quantitative Precipitation Information (AQPI) project, this study explores the potential of using the AQPI gap-filling radar network for streamflow simulation of selected storm events in the San Francisco Bay Area under a WRF-Hydro modeling system. Two types of watersheds including natural and human-affected among the most flood-prone region of the Bay Area are investigated. Based on the high-resolution AQPI X-band radar rainfall estimates, three basic routing configurations, including Grid, Reach, and National Water Model (NWM), are used to quantify the impact of different model physics options on the simulated streamflow. It is found that the NWM performs better in terms of reproducing streamflow volumes and hydrograph shapes than the other routing configurations when reservoirs exist in the watershed. Additionally, the AQPI X-band radar rainfall estimates (without gauge correction) provide reasonable streamflow simulations, and they show better performance in reproducing the hydrograph peaks compared with the gauge-corrected rainfall estimates based on the operational S-band Next Generation Weather Radar network. Also, sensitivity test reveals that surficial conditions have a significant influence on the streamflow simulation during the storm: the discharge increases to a higher level as the infiltration factor (REFKDT) decreases, and its peak goes down and lags as surface roughness coefficient (Mann) increases. The time delay analysis of precipitation input on the streamflow at the two outfalls of the surveyed watersheds further demonstrates the link between AQPI gap-filling radar observations and streamflow changes in this urban region.

57. A Vector-Based River Routing Model for Earth System Models: Parallelization and Global Applications

Authors: Naoki Mizukami (National Center for Atmospheric Research); Martyn P. Clark (Department of Geography and Planning University of Saskatchewan Saskatoon SK Canada, Centre for Hydrology University of Saskatchewan Canmore AB Canada); Shervan Gharari (Centre for Hydrology University of Saskatchewan Canmore AB Canada); Erik Kluzek (National Center for Atmospheric Research); Ming Pan (Scripps Institution of Oceanography / University of California San Diego, Princeton University); Peirong Lin (Princeton University, Peking University); Hylke E. Beck (GloH2O Almere Netherlands); Dai Yamazaki (University of Tokyo)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ms002434

URL: https://doi.org/10.1029/2020ms002434

Publication Date: 2021-05-14

Volume/Issue: Volume 13, Issue 6

References: 42

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: river, Earth System modeling, parallel computing

Funding: National Center for Atmospheric Research, National Aeronautics and Space Administration, National Science Foundation

Open Access: N/A

Abstract: A vector-river network explicitly uses realistic geometries of river reaches and catchments for spatial discretization in a river model. This enables improving the accuracy of the physical properties of the modeled river system, compared to a gridded river network that has been used in Earth System Models. With a finer-scale river network, resolving smaller-scale river reaches, there is a need for efficient methods to route streamflow and its constituents throughout the river network. The purpose of this study is twofold: (1) develop a new method to decompose river networks into hydrologically independent tributary domains, where routing computations can be performed in parallel; and (2) perform global river routing simulations with two global river networks, with different scales, to examine the computational efficiency and the differences in discharge simulations at various temporal scales. The new parallelization method uses a hierarchical decomposition strategy, where each decomposed tributary is further decomposed into many sub-tributary domains, enabling hybrid parallel computing. This parallelization scheme has excellent computational scaling for the global domain where it is straightforward to distribute computations across many independent river basins. However, parallel computing for a single large basin remains challenging. The global routing experiments show that the scale of the vector-river network has less impact on the discharge simulations than the runoff input that is generated by the combination of land surface model and meteorological forcing. The scale of vector-river networks needs to consider the scale of local hydrologic features such as lakes that are to be resolved in the network.

58. Global riverine theoretical hydrokinetic resource assessment

Authors: Michael Ridgill; Simon P. Neill; Matt J. Lewis; Peter E. Robins; Sopan D. Patil

Journal: Renewable Energy

Publisher: Elsevier BV

DOI: 10.1016/j.renene.2021.04.109

URL: https://doi.org/10.1016/j.renene.2021.04.109

Publication Date: 2021-04-27

Volume/Issue: Volume 174

References: 77

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

59. On the discretization of river networks for large scale hydrologic-hydrodynamic models

Authors: Fernando Mainardi Fan (Universidade Federal do Rio Grande do Sul, Brasil); Vinícius Alencar Siqueira (Universidade Federal do Rio Grande do Sul, Brasil); Ayan Santos Fleischmann (Universidade Federal do Rio Grande do Sul, Brasil); João Paulo Fialho Brêda (Universidade Federal do Rio Grande do Sul, Brasil); Rodrigo Cauduro Dias de Paiva (Universidade Federal do Rio Grande do Sul, Brasil); Paulo Rógenes Monteiro Pontes (Instituto Tecnológico Vale, Brasil); Walter Collischonn (Universidade Federal do Rio Grande do Sul, Brasil)

Journal: RBRH

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/2318-0331.262120200070

URL: https://doi.org/10.1590/2318-0331.262120200070

Publication Date: 2021-04-02

Volume/Issue: Volume 26

References: 102

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: The discretization of river networks is a critical step for computing flow routing in hydrological models. However, when it comes to more complex hydrologic-hydrodynamic models, adaptations in the spatial representation of model calculation units are further required to allow cost-effective simulations, especially for large scale applications. The objective of this paper is to assess the impacts of river discretization on simulated discharge, water levels and numerical stability of a catchment-based hydrologic-hydrodynamic model, using a fixed river length (Δx) segmentation method. The case study was the Purus river basin, a sub-basin of the Amazon, which covers an area that accounts for rapid response upstream reaches to downstream floodplain rivers. Results indicate that the maximum and minimum discharges are less affected by the adopted Δx (reach-length), whereas water levels are more influenced by this selection. It is showed that for the explicit local inertial one-dimensional routing, Δx and the alpha parameter of CFL (Courant-Friedrichs-Lewy) condition must be carefully chosen to avoid mass balance errors. Additionally, a simple Froude number-based flow limiter to avoid numerical issues is proposed and tested.

60. TRITON: A Multi-GPU open source 2D hydrodynamic flood model

Authors: M. Morales-Hernández; Md B. Sharif; A. Kalyanapu; S.K. Ghafoor; T.T. Dullo; S. Gangrade; S.-C. Kao; M.R. Norman; K.J. Evans

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2021.105034

URL: https://doi.org/10.1016/j.envsoft.2021.105034

Publication Date: 2021-04-02

Volume/Issue: Volume 141

References: 66

Cited by: 76 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

61. Simulation of Hurricane Harvey flood event through coupled hydrologic-hydraulic models: Challenges and next steps

Authors: Tigstu T. Dullo (Department of Civil and Environmental Engineering Tennessee Technological University Cookeville Tennessee USA); Sudershan Gangrade (The Bredesen Center University of Tennessee Knoxville Tennessee USA, Oak Ridge National Laboratory); Mario Morales-Hernández (Oak Ridge National Laboratory); Md Bulbul Sharif (Department of Computer Science Tennessee Technological University Cookeville Tennessee USA); Shih-Chieh Kao (The Bredesen Center University of Tennessee Knoxville Tennessee USA, Oak Ridge National Laboratory); Alfred J. Kalyanapu (Department of Civil and Environmental Engineering Tennessee Technological University Cookeville Tennessee USA); Sheikh Ghafoor (Department of Computer Science Tennessee Technological University Cookeville Tennessee USA); Katherine J. Evans (The Bredesen Center University of Tennessee Knoxville Tennessee USA, Oak Ridge National Laboratory)

Journal: Journal of Flood Risk Management

Publisher: Wiley

DOI: 10.1111/jfr3.12716

URL: https://doi.org/10.1111/jfr3.12716

Publication Date: 2021-03-16

Volume/Issue: Volume 14, Issue 3

References: 76

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: Using the 2017 Hurricane Harvey flood event as a test case, this study set up a series of sensitivity analyses to highlight three challenges associated with large-scale flood inundation modeling, including (a) model parameterization, (b) errors in digital elevation models, and (c) effects of reservoir retention. Driven by radar-based hourly rainfall data, a series of hydrologic-hydraulic models including the VIC hydrologic model, RAPID routing model, and Flood2D-GPU hydrodynamic model are set up over Harris County, Texas, to simulate flood inundation and hazards. The results demonstrate the importance of hydrologic parameters in improving flood modeling. For a large flood event such as Hurricane Harvey, the effect of the initial water depths is insignificant. The Manning's n values may increase the peak water depth by ~1%, the flood extents by 65km2, and the high danger zone by ~6%. On the contrary, the bathymetry correction factors may reduce the flood extent by ~1.4% and the high-danger zone by ~4%. Reducing the reservoir storage capacity to 1% may increase the flood extent by ~4% and the high-danger zone by ~17%. This study may provide supporting information to guide and prioritize the development of future high-performance computing hydrodynamic large-scale flood simulations.

62. Combining Optical Remote Sensing, McFLI Discharge Estimation, Global Hydrologic Modeling, and Data Assimilation to Improve Daily Discharge Estimates Across an Entire Large Watershed

Authors: Yuta Ishitsuka (University of Massachusetts Amherst); Colin J. Gleason (University of Massachusetts Amherst); Mark W. Hagemann (Ohio State University, University of Massachusetts Amherst); Edward Beighley (Department of Civil and Environmental Engineering Northeastern University Boston MA USA); George H. Allen (Department of Geological Sciences University of North Carolina-Chapel Hill Chapel Hill NC USA, Department of Geography Texas A&M; University College Station TX USA); Dongmei Feng (University of Massachusetts Amherst); Peirong Lin (Princeton University); Ming Pan (Princeton University); Konstantinos Andreadis (University of Massachusetts Amherst); Tamlin M. Pavelsky (Department of Geological Sciences University of North Carolina-Chapel Hill Chapel Hill NC USA)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020wr027794

URL: https://doi.org/10.1029/2020wr027794

Publication Date: 2020-12-07

Volume/Issue: Volume 57, Issue 3

References: 92

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: data assimilation, remote sensing, global hydrologic modeling, river discharge, Google Earth Engine (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Remote sensing has gained attention as a novel source of primary information for estimating river discharge, and the Mass-conserved Flow Law Inversion (McFLI) approach has successfully estimated river discharge in ungauged basins solely from optical satellite data. However, McFLI currently suffers from two major drawbacks: (1) existing optical satellites lead to temporally and spatially sparse discharge estimates and (2) because of the assumptions required, McFLI cannot guarantee downstream flow continuity. Hydrological modeling has neither drawback, yet model accuracy is frequently limited by a lack of discharge observations. We therefore combine McFLI and models in a data assimilation framework applicable globally. We establish a daily "ungauged" baseline model for 28,998 reaches of the Missouri river basin forced by recently published global runoff data, which we do not calibrate. We estimate discharge via McFLI using ~1 million width measurements made from 12,000 Landsat scenes and assimilate McFLI into the model before validating at 403 USGS gauges. Results show that assimilated discharges did not impair already accurate baseline flows and achieved median improvements of 28% normalized root mean square error, 0.50 Nash-Sutcliffe efficiency (NSE), and 0.23 Kling-Gupta efficiency where baseline performance was poor (defined as baseline negative NSE, 225/403 reaches). We ultimately improved flows at 92% of these originally

poorly modeled gauges, even though Landsat images only provide McFLI discharges at 1.5% of reaches and 26% of simulated days. Our results suggest that the combination of McFLI and state-of-the-art hydrology models can improve flow estimations in ungauged basins globally.

63. Improving flood simulation capability of the WRF-Hydro-RAPID model using a multi-source precipitation merging method

Authors: Lijun Chao; Ke Zhang; Zong-Liang Yang; Jingfeng Wang; Peirong Lin; Jingjing Liang; Zhijia Li; Zhao Gu

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2020.125814

URL: https://doi.org/10.1016/j.jhydrol.2020.125814

Publication Date: 2020-12-01

Volume/Issue: Volume 592

References: 63

Cited by: 42 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Six Talent Peaks Project in Jiangsu Province, China Postdoctoral Science Foundation, Fundamental Research Funds for the Central Universities (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

64. Analyzing future extreme floods for the Mississippi River Basin

Authors: S.E. Lytle; A.A. Tavakoly; E. Yeates; J. Lewis

Journal: River Flow 2020

Publisher: CRC Press

DOI: 10.1201/b22619-288

URL: https://doi.org/10.1201/b22619-288

Publication Date: 2020-10-29

Volume/Issue: N/A

References: 1

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

65. Two-way coupling between the sub-grid land surface and rivernetworks in Earth system models

Authors: Nathaniel W. Chaney; Laura Torres-Rojas; Noemi Vergopolan; Colby K. Fisher

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2020-291

URL: https://doi.org/10.5194/gmd-2020-291

Publication Date: 2020-10-28

Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Unstructured grid

Funding: Climate Program Office

Open Access: N/A

Abstract: . Over the past decade, there has been appreciable progress towards modeling the water, energy, and carbon cycles at field-scales (10-100 m) over continental to global extents. One such approach, named HydroBlocks, accomplishes this task while maintaining computational efficiency via sub-grid tiles, or Hydrologic Response Units (HRUs), learned via a hierarchical clustering approach from available global high-resolution environmental data. However, until now, there has yet to be a macroscale river routing approach that is able to leverage HydroBlocks' approach to sub-grid heterogeneity, thus limiting the added value of field-scale land surface modeling in Earth System Models (e.g., riparian zone dynamics, irrigation from surface water, and interactive floodplains). This paper introduces a novel dynamic river routing scheme in HydroBlocks that is intertwined with the modeled field-scale land surface heterogeneity. The primary features of the routing scheme include: 1) the fine-scale river network of each macroscale grid cell's is derived from very high resolution (

66. Assimilation of future SWOT-based river elevations, surface extent observations and discharge estimations into uncertain global hydrological models

Authors: Sly Wongchuig-Correa; Rodrigo Cauduro Dias de Paiva; Sylvain Biancamaria; Walter Collischonn

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2020.125473

URL: https://doi.org/10.1016/j.jhydrol.2020.125473

Publication Date: 2020-08-31

Volume/Issue: Volume 590

References: 118

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Coordenação de Aperfeiçoamento de Pessoal de Nível Superior

License: N/A

Open Access: N/A

Abstract: N/A

67. Subwatershed-based lake and river routing products for hydrologic and land surface models applied over Canada

Authors: Ming Han (University of Waterloo); Juliane Mai (University of Waterloo); Bryan A. Tolson (University of Waterloo); James R. Craig (University of Waterloo); Étienne Gaborit (Environment and Climate Change Canada); Hongli Liu (University of Waterloo); Konhee Lee (University of Waterloo)

Journal: Canadian Water Resources Journal / Revue canadienne des ressources hydriques

Publisher: Informa UK Limited

DOI: 10.1080/07011784.2020.1772116

URL: https://doi.org/10.1080/07011784.2020.1772116

Publication Date: 2020-06-12

Volume/Issue: Volume 45, Issue 3

References: 35

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

68. Pluri-annual Water Budget on the Seine Basin: Past, Current and Future Trends

Authors: Nicolas Flipo; Nicolas Gallois; Baptiste Labarthe; Fulvia Baratelli; Pascal Viennot; Jonathan Schuite; Agnès Rivière; Rémy Bonnet; Julien Boé

Journal: The Handbook of Environmental Chemistry

Publisher: Springer International Publishing

DOI: 10.1007/698_2019_392

URL: https://doi.org/10.1007/698_2019_392

Publication Date: 2020-06-02

Volume/Issue: N/A

References: 112

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Water budget, CaWaQS, Climate change, Groundwater, Surface water (and 4 more)

Funding: N/A

Open Access: N/A

Abstract: The trajectory of the Seine basin water resources is rebuilt from the early 1900s to the 2000s before being projected to the end of the twenty-first century. In the first part, the long-term hydrological data of the Paris gauging stations are analysed beginning in 1885, highlighting the effect of anthropogenic water management on the Seine River discharge. Then a detailed water budget of the Seine basin is proposed. It quantifies for the first time the water exchanges between aquifer units and the effect of water withdrawals on river-aquifer exchanges. Using this model, the trajectory of the system is evaluated based on a downscaled climate reanalysis of the twentieth century and a reconstruction of the land use in the early 1900s, as well as the choice of a climate projection which favours the model that best reproduces the low frequency of precipitation. The trajectory is synthesised as average regimes, revealing a relative stability of the hydrosystem up to the present, and drastic changes in the discharge regime in the future, especially concerning the decreased amount of low flow and its increased duration. These expected changes will require the definition of an adaptation strategy even though they are rather limited in the Seine basin when compared to other French regions.

69. Comparison of generalized non-data-driven lake and reservoir routing models for global-scale hydrologic forecasting of reservoir outflow at diurnal time steps

Authors: Joseph L. Gutenson; Ahmad A. Tavakoly; Mark D. Wahl; Michael L. Follum

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-24-2711-2020

URL: https://doi.org/10.5194/hess-24-2711-2020

Publication Date: 2020-05-27

Volume/Issue: Volume 24, Issue 5

References: 47

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . Large-scale hydrologic forecasts should account for attenuationthrough lakes and reservoirs when flow regulation is present. Globallygeneralized methods for approximating outflow are required but must contendwith operational complexity and a dearth of information on damcharacteristics at global spatial scales. There is currently no consensus onthe best approach for approximating reservoir release rates in large spatialscale hydrologic forecasting, particularly at diurnal time steps. Thisresearch compares two parsimonious reservoir routing methods at daily steps: Döll et al. (2003) and Hanasaki et al. (2006). These reservoir routingmethods have been previously implemented in large-scale hydrologic modelingapplications and have been typically evaluated seasonally. These routingmethods are compared across 60 reservoirs operated by the U.S. Army Corps ofEngineers. The authors vary empirical coefficients for both reservoirrouting methods as part of a sensitivity analysis. The method proposed byDöll et al. (2003) outperformed that presented by Hanasaki et al. (2006)at a daily time step and improved model skill over most run-of-the-riverconditions. The temporal resolution of the model influences modelperformances. The optimal model coefficients varied across the reservoirs inthis study and model performance fluctuates between wet years and dry years,and for different configurations such as dams in series. Overall, the methodproposed by Döll et al. (2003) could enhance large-scale hydrologicforecasting, but can be subject to instability under certain conditions.

70. Global Estimates of Reach-Level Bankfull River Width Leveraging Big Data Geospatial Analysis

Authors: Peirong Lin (Princeton University); Ming Pan (Princeton University); George H. Allen (Department of Geography Texas A&M; University College Station TX USA); Renato Prata de Frasson (Ohio State University); Zhenzhong Zeng (Princeton University, Southern University of Science and Technology); Dai Yamazaki (University of Tokyo); Eric F. Wood (Princeton University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl086405

URL: https://doi.org/10.1029/2019gl086405

Publication Date: 2020-04-02

Volume/Issue: Volume 47, Issue 7

References: 69

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: global bankfull river width, channel geometry, machine learning, hydrodynamic modeling

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Recent progress in remote sensing has snapshotted unprecedented numbers of river planform geometry, providing opportunity to revisit the oversimplified channel shape parameterizations in global hydrologic models. This study leveraged two recent Landsat-derived global river width databases and created a reach-level width dataset to measure the validity of model parameterizations at ~1.6 million kilometers of rivers in length. By showing state-of-the-art parameterization schemes only capture 30-40% of the width variance globally, we developed a machine learning (ML) approach surveying 16 environmental covariates, which considerably improved the predictive power (R2 = 0.81 and 0.77 for two testing cases). Beyond the commonly discussed upstream basin conditions, ML revealed that local physiographic factors and human interference are also important covariates for width variability. Finally, we applied the ML model to estimate bankfull river width, creating a new reach-level dataset for use in global hydrodynamic modeling.

71. Improved accuracy and efficiency of flood inundation mapping of low-, medium-, and high-flow events using the AutoRoute model

Authors: Michael L. Follum; Ricardo Vera; Ahmad A. Tavakoly; Joseph L. Gutenson

Journal: Natural Hazards and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/nhess-20-625-2020

URL: https://doi.org/10.5194/nhess-20-625-2020

Publication Date: 2020-02-28

Volume/Issue: Volume 20, Issue 2

References: 45

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . This article presents improvements and the development of a postprocessing module for the regional-scale flood mapping tool, AutoRoute. The accuracy of this model to simulate low-, medium-, and high-flow-rate scenarios is demonstrated at seven test sites within the US. AutoRoute is one of the tools used to create high-resolution flood inundation maps at regional to continental scales, but it has previously only been tested using extreme flood events. Modifications to the AutoRoute model and postprocessing scripts are shown to improve accuracy (e.g., average F value increase of 17.5 % for low-flow events) and computational efficiency (simulation time reduced by over 40 %) when compared to previous versions. Although flood inundation results for low-flow events are shown to be comparable with published values (average F value of 63.3 %), the model results tend to be overestimated, especially in flatter terrain. Higher-flow scenarios tend to be more accurately simulated (average F value of 77.5 %). With improved computational efficiency and the enhanced ability to simulate both low- and high-flow scenarios, the AutoRoute model may be well suited to provide first-order estimates of flooding within an operational, regional- to continental-scale hydrologic modeling framework.

72. The AquiFR hydrometeorological modelling platform as a tool for improving groundwater resource monitoring over France: evaluation over a 60-year period

Authors: Jean-Pierre Vergnes; Nicolas Roux; Florence Habets; Philippe Ackerer; Nadia Amraoui; François Besson; Yvan Caballero; Quentin Courtois; Jean-Raynald de Dreuzy; Pierre Etchevers; Nicolas Gallois; Delphine J. Leroux; Laurent Longuevergne; Patrick Le Moigne; Thierry Morel; Simon Munier; Fabienne Regimbeau; Dominique Thiéry; Pascal Viennot

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-24-633-2020

URL: https://doi.org/10.5194/hess-24-633-2020

Publication Date: 2020-02-13

Volume/Issue: Volume 24, Issue 2

References: 85

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . The new AquiFR hydrometeorological modelling platform was developed to provide short-to-long-term forecasts for groundwater resource management in France. This study aims to describe and assess this new tool over a long period of 60 years. This platform gathers in a single numerical tool several hydrogeological models covering much of the French metropolitan area. A total of 11 aquifer systems are simulated through spatially distributedmodels using either the MARTHE (Modélisation d'Aquifères avec un maillage Rectangulaire, Transport et HydrodynamiquE; Modelling Aquifers with Rectangular cells, Transport and Hydrodynamics) groundwater modelling software programme or theEauDyssée hydrogeological platform. A total of 23 karstic systems aresimulated by a lumped reservoir approach using the EROS (Ensemble de Rivières Organisés en Sous-bassins; set of rivers organized in sub-basins) software programme. AquiFRcomputes the groundwater level, the groundwater-surface-water exchanges and the river flows. A simulation covering a 60-year period from 1958 to 2018 is achieved in order to evaluate the performance of this platform. The 8 km resolution SAFRAN (Système d'Analyse Fournissant des Renseignements Adaptés à la Nivologie) meteorological analysis provides the atmospheric variables needed by the SURFEX (SURFace EXternalisée) land surface model in order to compute surface runoff and groundwater recharge used by the hydrogeological models. The assessment is based on more than 600 piezometers and more than 300 gauging stations corresponding to simulated rivers and outlets of karstic systems. For the simulated piezometric heads, 42 % and 60 % of the absolute biases are lower than 2 and 4 m respectively. The standardized piezometric level index (SPLI) was computed to assess the ability of AquiFR to identify extreme events such as groundwater floods or droughts in the long-term simulation over a set of piezometers used for groundwater resource management. A total of 56 % of the Nash-Sutcliffe efficiency (NSE; Ef) coefficient calculations between the observed and simulated SPLI time series are greater than 0.5. The quality of the results makes it possible to consider using the platform for real-time

monitoring and seasonal forecasts of groundwater resources as well as for climate change impact assessments.

73. Underlying Fundamentals of Kalman Filtering for River Network Modeling

Authors: Charlotte M. Emery (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Konstantinos M. Andreadis (University of Massachusetts Amherst); Michael J. Turmon (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jonathan M. Hobbs (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ming Pan (Princeton University); James S. Famiglietti (d Global Institute for Water Security, University of Saskatchewan, Saskatoon, Saskatchewan, Canada); Edward Beighley (e Department of Civil and Environment Engineering, Northeastern University, Boston, Massachusetts); Matthew Rodell (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm-d-19-0084.1

URL: https://doi.org/10.1175/jhm-d-19-0084.1

Publication Date: 2020-02-03

Volume/Issue: Volume 21, Issue 3

References: 106

Cited by: 11 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: SWOT Science Team, NASA ROSES - Terrestrial Hydrology Program

License: N/A

Open Access: N/A

Abstract: The grand challenge of producing hydrometeorological estimates every time and everywhere has motivated the fusion of sparse observations with dense numerical models, with a particular interest on discharge in river modeling. Ensemble methods are largely preferred as they enable the estimation of error properties, but at the expense of computational load and generally with underestimations. These imperfect stochastic estimates motivate the use of correction methods, that is, error localization and inflation, although the physical justifications for their optimality are limited. The purpose of this study is to use one of the simplest forms of data assimilation when applied to river modeling and reveal the underlying mechanisms impacting its performance. Our framework based on assimilating daily averaged in situ discharge measurements to correct daily averaged runoff was tested over a 4-yr case study of two rivers in Texas. Results show that under optimal conditions of inflation and localization, discharge simulations are consistently improved such that the mean values of Nash-Sutcliffe efficiency are enhanced from -11.32 to 0.55 at observed gauges and from -12.24 to -1.10 at validation gauges. Yet, parameters controlling the inflation and the localization have a large impact on the performance. Further investigations of these sensitivities showed that optimal inflation occurs when compensating exactly for discrepancies in the magnitude of errors while optimal localization matches the distance traveled during one assimilation window. These results may be applicable to more advanced data

assimilation methods as well as for larger applications motivated by upcoming river-observing satellite missions, such as NASA's Surface Water and Ocean Topography mission.

74. Towards a large-scale locally relevant flood inundation modeling framework using SWAT and LISFLOOD-FP

Authors: Adnan Rajib; Zhu Liu; Venkatesh Merwade; Ahmad A. Tavakoly; Michael L. Follum

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2019.124406

URL: https://doi.org/10.1016/j.jhydrol.2019.124406

Publication Date: 2019-11-28

Volume/Issue: Volume 581

References: 104

Cited by: 100 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

75. Evaluation of Available Global Runoff Datasets Through a River Model in Support of Transboundary Water Management in South and Southeast Asia

Authors: Md. Safat Sikder; Cédric H. David; George H. Allen; Xiaohui Qiao; E. James Nelson; Mir A. Matin

Journal: Frontiers in Environmental Science

Publisher: Frontiers Media SA

DOI: 10.3389/fenvs.2019.00171

URL: https://doi.org/10.3389/fenvs.2019.00171

Publication Date: 2019-10-31

Volume/Issue: Volume 7

References: 63

Cited by: 20 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Abstract: N/A

76. Hydrologic Modeling as a Service (HMaaS): A New Approach to Address Hydroinformatic Challenges in Developing Countries

Authors: Michael A. Souffront Alcantara; E. James Nelson; Kiran Shakya; Christopher Edwards; Wade Roberts; Corey Krewson; Daniel P. Ames; Norman L. Jones; Angelica Gutierrez

Journal: Frontiers in Environmental Science

Publisher: Frontiers Media SA

DOI: 10.3389/fenvs.2019.00158

URL: https://doi.org/10.3389/fenvs.2019.00158

Publication Date: 2019-10-23

Volume/Issue: Volume 7

References: 37

Cited by: 32 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: N/A

77. Mississippi River and Tributaries future flood conditions

Authors: James Lewis; Ahmad Tavakoly; Charles Martin; Christine Moore

Journal: Unknown Journal

Publisher: Engineer Research and Development Center (U.S.)

DOI: 10.21079/11681/34206

URL: https://doi.org/10.21079/11681/34206

Publication Date: 2019-09-26

Volume/Issue: N/A

References: N/A

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: River flood

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

78. In Quest of Calibration Density and Consistency in Hydrologic Modeling: Distributed Parameter Calibration against Streamflow Characteristics

Authors: Yuan Yang (Princeton University, Tsinghua University); Ming Pan (Princeton University); Hylke E. Beck (Princeton University); Colby K. Fisher (Princeton University); R. Edward Beighley (Department of Civil and Environmental Engineering Northeastern University Boston Massachusetts USA); Shih-Chieh Kao (Oak Ridge National Laboratory); Yang Hong (University of Oklahoma, Tsinghua University); Eric F. Wood (Princeton University)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018wr024178

URL: https://doi.org/10.1029/2018wr024178

Publication Date: 2019-08-31

Volume/Issue: Volume 55, Issue 9

References: 88

Cited by: 53 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: Conventional basin-by-basin approaches to calibrate hydrologic models are limited to gauged basins and typically result in spatially discontinuous parameter fields. Moreover, the consequent low calibration density in space falls seriously behind the need from present-day applications like high resolution river hydrodynamic modeling. In this study we calibrated three key parameters of the Variable Infiltration Capacity (VIC) model at every 1/8 degrees grid-cell using machine learning-based maps of four streamflow characteristics for the conterminous United States (CONUS), with a total of 52,663 grid-cells. This new calibration approach, as an alternative to parameter regionalization, applied to ungauged regions too. A key difference made here is that we tried to regionalize physical variables (streamflow characteristics) instead of model parameters whose behavior may often be less well understood. The resulting parameter fields no longer presented any spatial discontinuities and the patterns corresponded well with climate characteristics, such as aridity and runoff ratio. The calibrated parameters were evaluated against observed streamflow from 704/648 (calibration/validation period) small-to-medium-sized catchments used to derive the streamflow characteristics, 3941/3809 (calibration/validation period) small-to-medium-sized catchments not used to derive the streamflow characteristics as well as five large basins. Comparisons indicated marked improvements in bias and Nash-Sutcliffe efficiency. Model performance was still poor in arid and semiarid regions, which is mostly due to both model structural and forcing deficiencies. Although the performance gain was limited by the relative small number of parameters to calibrate, the study and results here served as a proof-of-concept for a new promising approach for fine-scale hydrologic model calibrations.

79. A systems approach to routing global gridded runoff through local high-resolution stream networks for flood early warning systems

Authors: Xiaohui Qiao; E. James Nelson; Daniel P. Ames; Zhiyu Li; Cédric H. David; Gustavious P. Williams; Wade Roberts; Jorge Luis Sánchez Lozano; Chris Edwards; Michael Souffront; Mir A. Matin

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2019.104501

URL: https://doi.org/10.1016/j.envsoft.2019.104501

Publication Date: 2019-08-09

Volume/Issue: Volume 120

References: 55

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: Flood forecasting|Flood warning

Funding: NASA ROSES SERVIR Applied Sciences Team Research, Jet Propulsion Laboratory, California Institute of Technology (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

80. Global Reconstruction of Naturalized River Flows at 2.94 Million Reaches

Authors: Peirong Lin (Princeton University); Ming Pan (Princeton University); Hylke E. Beck (Princeton University); Yuan Yang (Princeton University, Tsinghua University); Dai Yamazaki (University of Tokyo); Renato Frasson (Ohio State University); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael Durand (Ohio State University); Tamlin M. Pavelsky (Department of Geological Sciences University of North Carolina at Chapel Hill Chapel Hill NC USA); George H. Allen (Department of Geography Texas A&M; University College Station TX USA); Colin J. Gleason (University of Massachusetts Amherst); Eric F. Wood (Princeton University)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019wr025287

URL: https://doi.org/10.1029/2019wr025287

Publication Date: 2019-07-24

Volume/Issue: Volume 55, Issue 8

References: 83

Cited by: 213 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Army Corps of Engineers, Jet Propulsion Laboratory, National Aeronautics and Space Administration

Open Access: N/A

Abstract: Spatiotemporally continuous global river discharge estimates across the full spectrum of stream orders are vital to a range of hydrologic applications, yet they remain poorly constrained. Here we present a carefully designed modeling effort (Variable Infiltration Capacity land surface model and Routing Application for Parallel computatIon of Discharge river routing model) to estimate global river discharge at very high resolutions. The precipitation forcing is from a recently published 0.1 degrees global product that optimally merged gauge-, reanalysis-, and satellite-based data. To constrain runoff simulations, we use a set of machine learning-derived, global runoff characteristics maps (i.e., runoff at various exceedance probability percentiles) for grid-by-grid model calibration and bias correction. To support spaceborne discharge studies, the river flowlines are defined at their true geometry and location as much as possible--approximately 2.94 million vector flowlines (median length 6.8 km) and unit catchments are derived from a high-accuracy global digital elevation model at 3-arcsec resolution (~90 m), which serves as the underlying hydrography for river routing. Our 35-year daily and monthly model simulations are evaluated against over 14,000 gauges globally. Among them, 35% (64%) have a percentage bias within +/-20% (+/-50%), and 29% (62%) have a monthly Kling-Gupta Efficiency >=0.6 (0.2), showing data robustness at the scale the model is assessed. This reconstructed discharge record can be used as a priori information for the Surface Water and Ocean Topography satellite mission's discharge product, thus named "Global Reach-level A priori Discharge Estimates for Surface Water and Ocean Topography". It can also be used in other hydrologic applications requiring spatially explicit estimates of global river flows.

81. Comparison of Generalized Non-Data-Driven Reservoir Routing Models for Global-Scale Hydrologic Modeling

Authors: Joseph L. Gutenson; Ahmad A. Tavakoly; Mark D. Wahl; Michael L. Follum

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/hess-2019-264

URL: https://doi.org/10.5194/hess-2019-264

Publication Date: 2019-07-04

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Hydrological modelling|Flow routing|Outflow|Scale model

Funding: N/A

Open Access: N/A

Abstract: . Large-scale hydrologic simulations should account for attenuation through lakes and reservoirs when flow regulation is present. Generalized methods for approximating outflow are required since reservoir operation is complex and specific real-time release information is typically unavailable at global scales. There is currently no consensus on the best approach for approximating reservoir release rates in large spatial scale hydrologic forecasting. This research compares two parsimonious reservoir routing methods previously implemented in large-scale hydrologic modeling applications, requiring minimal data so as not to limit their usage. The methods considered are those proposed by Döll et al. (2003) and Hanasaki et al. (2006). This paper compares the two methodologies across 60 reservoirs operated from 2006-2012 by the U.S. Army Corps of Engineers. The authors vary empirical coefficients for both reservoir routing methods as part of a sensitivity analysis. The Döll method generally outperformed the Hanasaki method at a daily time step, improving model skill in most cases beyond run-of-the-river conditions. The temporal resolution of the model influences performance. The optimal model coefficients varied across the reservoirs in this study and model performance fluctuates between wet years and dry years, and for different configurations such as dams in series. Overall, the Döll and Hanasaki Methods could enhance large scale hydrologic forecasting, but can be subject to instability under certain conditions.

82. The multiscale routing model mRM v1.0: simple river routing at resolutions from 1 to 50 km

Authors: Stephan Thober; Matthias Cuntz; Matthias Kelbling; Rohini Kumar; Juliane Mai; Luis Samaniego

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-2501-2019

URL: https://doi.org/10.5194/gmd-12-2501-2019

Publication Date: 2019-06-28

Volume/Issue: Volume 12, Issue 6

References: 77

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung, Agence Nationale de la Recherche

Abstract: . Routing streamflow through a river network is a fundamentalrequirement to verify lateral water fluxes simulated by hydrologicand land surface models. River routing is performed at diverseresolutions ranging from few kilometres to 1■. The presentedmultiscale routing model mRM calculates streamflow at diversespatial and temporal resolutions. mRM solves the kinematic waveequation using a finite difference scheme. An adaptive time steppingscheme fulfilling a numerical stability criterion is introduced inthis study and compared against the original parameterisation of mRMthat has been developed within the mesoscale hydrologic model(mHM). mRM requires a high-resolution river network, which isupscaled internally to the desired spatial resolution. The user canchange the spatial resolution by simply changing a single number inthe configuration file without any further adjustments of the inputdata. The performance of mRM is investigated on two datasets: ahigh-resolution German dataset and a slightly lower resolvedEuropean dataset. The adaptive time stepping scheme within mRM showsa remarkable scalability compared to its predecessor. MedianKling-Gupta efficiencies change less than 3 % when the modelparameterisation is transferred from 3 to 48 km resolution. mRM alsoexhibits seamless scalability in time, providing similar resultswhen forced with hourly and daily runoff. The streamflow calculatedover the Danube catchment by the regional climate model REMO coupledto mRM reveals that the 50 km simulation shows a smaller bias withrespect to observations than the simulation at 12 kmresolution. The mRM source code is freely available and highlymodular, facilitating easy internal coupling in existing Earthsystem models.

83. Flood Inundation Mapping of Low-, Medium-, and High-Flow EventsUsing the AutoRoute Model

Authors: Michael L. Follum; Ahmad A. Tavakoly; Joseph L. Gutenson; Ricardo Vera

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/nhess-2019-180

URL: https://doi.org/10.5194/nhess-2019-180

Publication Date: 2019-06-17

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . This article presents improvements and development of a post-processing module for the regional scale flood mapping tool, AutoRoute. The accuracy of this model to simulate low, medium, and high flow rate scenarios is demonstrated at seven test sites within the U.S. AutoRoute is one of the tools used to create high-resolution flood inundation maps at regional- to continental-scales. The model has previously only been tested using extreme flood events. In this article flood inundation results for low-flow events are shown to be accurate (average F value of 63.3 %) but tend to be overestimated, especially in flatter terrain. Higher-flow scenarios tend to be more accurately simulated (average F value of 77.5 %). Additionally, modifications to the AutoRoute model and post-processing scripts are shown to improve computational efficiency (i.e. simulation time) by over 40 % when compared to previous versions. With improved computational efficiency and the ability to accurately simulate both low and high flow scenarios the AutoRoute model may be well suited to provide first-order estimates of flooding within an operational, regional- to continental-scale hydrologic modelling framework.

84. Analytical Propagation of Runoff Uncertainty Into Discharge Uncertainty Through a Large River Network

Authors: Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jonathan M. Hobbs (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael J. Turmon (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charlotte M. Emery (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); James S. Famiglietti (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Now at Global Institute for Water Security University of Saskatchewan Saskatoon Saskatchewan Canada)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083342

URL: https://doi.org/10.1029/2019gl083342

Publication Date: 2019-05-30

Volume/Issue: Volume 46, Issue 14

References: 64

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: runoff, land, river, uncertainty, discharge (and 1 more)

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration, National Aeronautics and Space Administration (and 1 more)

License: N/A

Open Access: N/A

Abstract: The transport of freshwater from continents to oceans through rivers has traditionally been estimated by routing runoff from land surface models within river models to obtain discharge. This paradigm imposes that errors are transferred from runoff to discharge, yet the analytical propagation of uncertainty from runoff to discharge has never been derived. Here we apply statistics to the continuity equation within a river network to derive two equations that propagate the mean and variance/covariance of runoff errors independently. We validate these equations in a case study of the rivers in the western United States and, for the first time, invert observed discharge errors for spatially distributed runoff errors. Our results suggest that the largest discharge error source is the joint variability of runoff errors across space, not the mean or amplitude of individual errors. Our findings significantly advance the science of error quantification in model-based estimates of river discharge.

85. Evaluation and uncertainty attribution of the simulated streamflow from NoahMP-RAPID over a high-altitude mountainous basin

Authors: Shu Wang; Hui Zheng; Peirong Lin; Xing Zheng; Zong-Liang Yang

Journal: Chinese Science Bulletin

Publisher: Science China Press., Co. Ltd.

DOI: 10.1360/n972018-00598

URL: https://doi.org/10.1360/n972018-00598

Publication Date: 2019-01-22

Volume/Issue: Volume 64, Issue 4

References: N/A

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

86. Assimilation of Synthetic SWOT River Depths in a Regional Hydrometeorological Model

Authors: Vincent Häfliger (CNRM-GAME, UMR 3589, Météo-France, CNRS, 31057 Toulouse, France, Meteorologisches Institut, Bonn Universität, 53113 Bonn, Germany); Eric Martin (CNRM-GAME, UMR 3589, Météo-France, CNRS, 31057 Toulouse, France, IRSTEA, UR RECOVER, 13100 Aix-en-Provence, France); Aaron Boone (CNRM-GAME, UMR 3589, Météo-France, CNRS, 31057 Toulouse, France); Sophie Ricci (CECI, CERFACS-CNRS, 42 avenue G. Coriolis, 31057 Toulouse, France); Sylvain Biancamaria (Institut de Recherche pour le Développement (IRD))

Journal: Water

Publisher: MDPI AG

DOI: 10.3390/w11010078

URL: https://doi.org/10.3390/w11010078

Publication Date: 2019-01-04

Volume/Issue: Volume 11, Issue 1

References: 41

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: The SWOT (Surface Water and Ocean Topography) mission, to be launched in 2021, will provide water surface elevations, slopes, and river width measurements for rivers wider than 100 m. In this study, synthetic SWOT data are assimilated in a regional hydrometeorological model in order to improve the dynamics of continental waters over the Garonne catchment, one of the major French catchments. The aim of this paper is to demonstrate that the sequential assimilation of SWOT-like river depths allows the correction of river bed roughness coefficients and thus simulated river depths. An extended Kalman filter is implemented and the data assimilation strategy was applied to four experiments of gradually increasing complexity regarding observation and model error over the 1995-2000 period. With respect to a "true" river state, assimilating river depths allows the proper retrieval of constant and spatially distributed roughness coefficients with a root mean square error of 1 m1/3 s-1, and the estimation of associated river depths. It was also shown that river depth differences can be assimilated, resulting in a higher root mean square error for roughness coefficients with respect to the true river state. Finally, the last experiment shows how one can take into account more realistic sources of SWOT error measurements, in particular the importance of the estimation of the tropospheric water content in the process.

87. ORCHIDEE-ROUTING: revising the river routing scheme using a high-resolution hydrological database

Authors: Trung Nguyen-Quang; Jan Polcher; Agnès Ducharne; Thomas Arsouze; Xudong Zhou; Ana Schneider; Lluís Fita

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-4965-2018

URL: https://doi.org/10.5194/gmd-11-4965-2018

Publication Date: 2018-12-07

Volume/Issue: Volume 11, Issue 12

References: 91

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: École Polytechnique, Université Paris-Saclay

Open Access: N/A

Abstract: . The river routing scheme (RRS) in the Organising Carbon and Hydrology in Dynamic Ecosystems (ORCHIDEE)land surface model is a valuable tool for closing the water cycle in acoupled environment and for validating the model performance. This studypresents a revision of the RRS of the ORCHIDEE model that aims to benefitfrom the high-resolution topography provided by the Hydrological data andmaps based on SHuttle Elevation Derivatives at multiple Scales (HydroSHEDS),which is processed to a resolution of approximately 1 km. Adapting a newalgorithm to construct river networks, the new RRS in ORCHIDEE allows for thepreservation of as much of the hydrological information from HydroSHEDS asthe user requires. The evaluation focuses on 12 rivers of contrasting sizeand climate which contribute freshwater to the Mediterranean Sea. First, thenumerical aspect of the new RRS is investigated, in order to identify thepractical configuration offering the best trade-off between computationalcost and simulation quality for ensuing validations. Second, the performanceof the new scheme is evaluated against observations at both monthly and dailytimescales. The new RRS satisfactorily captures the seasonal variability ofriver discharge, although important biases stem from the water budgetsimulated by the ORCHIDEE model. The results highlight that realisticstreamflow simulations require accurate precipitation forcing data and aprecise river catchment description over a wide range of scales, as permittedby the new RRS. Detailed analyses at the daily timescale show the promisingperformance of this high-resolution RRS with respect to replicating riverflow variation at various frequencies. Furthermore, this RRS may alsoeventually be well adapted for further developments in the ORCHIDEE landsurface model to assess anthropogenic impacts on river processes (e.g.damming for irrigation operation).

88. Empirical assessment of flood wave celerity-discharge relationships at local and reach scales

Authors: Aline Meyer (Instituto de Pesquisas Hidráulicas, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil); Ayan Santos Fleischmann (Instituto de Pesquisas Hidráulicas, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil); Walter Collischonn (Instituto de Pesquisas Hidráulicas, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil); Rodrigo Paiva (Instituto de Pesquisas Hidráulicas, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil); Pedro Jardim (Instituto de Pesquisas Hidráulicas, Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil)

Journal: Hydrological Sciences Journal

Publisher: Informa UK Limited

DOI: 10.1080/02626667.2018.1557336

URL: https://doi.org/10.1080/02626667.2018.1557336

Publication Date: 2018-12-07

Volume/Issue: Volume 63, Issue 15-16

References: 49

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Pró-Reitoria de Pesquisa, Universidade Federal do Rio Grande do Sul

License: N/A

Open Access: N/A

Abstract: N/A

89. An integrated framework to model nitrate contaminants with interactions of agriculture, groundwater, and surface water at regional scales: The STICS-EauDyssée coupled models applied over the Seine River Basin

Authors: Ahmad A. Tavakoly; Florence Habets; Firas Saleh; Zong-Liang Yang; Cyril Bourgeois; David R. Maidment

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2018.11.061

URL: https://doi.org/10.1016/j.jhydrol.2018.11.061

Publication Date: 2018-11-26

Volume/Issue: Volume 568

References: 94

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA, USACE

License: N/A

Open Access: N/A

Abstract: N/A

90. High-spatial-resolution streamflow estimation at ungauged river sites or gauged sites with missing data using the National Hydrography Dataset (NHD) and U.S. Geological Survey (USGS) streamflow data

Authors: Dong Ha Kim

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2018.08.074

URL: https://doi.org/10.1016/j.jhydrol.2018.08.074

Publication Date: 2018-09-01

Volume/Issue: Volume 565

References: 72

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

91. Insights into Hydrometeorological Factors Constraining Flood Prediction Skill during the May and October 2015 Texas Hill Country Flood Events

Authors: Peirong Lin (University of Texas at Austin); Larry J. Hopper (NOAA/NWS Austin/San Antonio Weather Forecast Office, New Braunfels, Texas); Zong-Liang Yang (Chinese Academy of Sciences, University of Texas at Austin); Mark Lenz (NOAA/NWS Austin/San Antonio Weather Forecast Office, New Braunfels, Texas); Jon W. Zeitler (NOAA/NWS Austin/San Antonio Weather Forecast Office, New Braunfels, Texas)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm-d-18-0038.1

URL: https://doi.org/10.1175/jhm-d-18-0038.1

Publication Date: 2018-07-25

Volume/Issue: Volume 19, Issue 8

References: 61

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: University Corporation for Atmospheric Research

License: N/A

Open Access: N/A

Abstract: This study evaluates the May and October 2015 flood prediction skill of a physically based model resembling the U.S. National Water Model (NWM) over the Texas Hill Country. It also investigates hydrometeorological factors that contributed to a record flood along the Blanco River at Wimberley (WMBT2) in May 2015. Using two radar-based quantitative precipitation estimation (QPE) products--Stage IV and Multi-Radar Multi-Sensor (MRMS)--it is shown that the event precipitation accuracy dominates the prediction skill, where the finer-resolution MRMS QPE mainly benefits basins with small drainage areas. Overall, the model exhibits good performance at gauges with fast flood response from causative rainfall and gauges that are not forecast points in the National Weather Service's Advanced Hydrometeorological Prediction System, showing great promise for forecasts, warnings, and emergency response. However, the model suffers from poor prediction skill over regions without rapid flood response and regions with human-altered flows, suggesting the need to revisit the channel routing algorithm and incorporate modules to represent human alterations. Two contrasting flood events at WMBT2 with similar meteorological characteristics are examined in greater detail, revealing that the location of intense rainfall combined with land physiographic features are key to the flood response differences. Model sensitivity tests further show the record flood peak could be better obtained by tuning the deep-layer soil wetness and the flow velocity field in the river network, which offers hydrometeorological insights into the causes and the complex nature of such a flood and why the model struggles to predict the record flood peak.

92. A CyberGIS Integration and Computation Framework for High-Resolution Continental-Scale Flood Inundation Mapping

Authors: Yan Y. Liu (University of Illinois); David R. Maidment (Center for Water and Environment University of Texas Austin Texas USA); David G. Tarboton (Utah State University); Xing Zheng (Department of Civil, Architectural and Environmental Engineering University of Texas Austin Texas USA); Shaowen Wang (University of Illinois)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12660

URL: https://doi.org/10.1111/1752-1688.12660

Publication Date: 2018-06-12

Volume/Issue: Volume 54, Issue 4

References: 36

Cited by: 64 publications

Language: en

Subjects: N/A

Keywords: Cyberinfrastructure

Funding: National Science Foundation, U.S. Geological Survey, U.S. Army Corps of Engineers

License: N/A

Open Access: N/A

Abstract: We present a Digital Elevation Model-based hydrologic analysis methodology for continental flood inundation mapping (CFIM), implemented as a cyberGIS scientific workflow in which a 1/3rd arc-second (10 m) height above nearest drainage (HAND) raster data for the conterminous United States (CONUS) was computed and employed for subsequent inundation mapping. A cyberGIS framework was developed to enable spatiotemporal integration and scalable computing of the entire inundation mapping process on a hybrid supercomputing architecture. The first 1/3rd arc-second CONUS HAND raster dataset was computed in 1.5 days on the cyberGIS Resourcing Open Geospatial Education and Research supercomputer. The inundation mapping process developed in our exploratory study couples HAND with National Water Model forecast data to enable near real-time inundation forecasts for CONUS. The computational performance of HAND and the inundation mapping process were profiled to gain insights into the computational characteristics in high-performance parallel computing scenarios. The establishment of the CFIM computational framework has broad and significant research implications that may lead to further development and improvement of flood inundation mapping methodologies.

93. Implementation of a vector-based river network routing scheme in the community WRF-Hydro modeling framework for flood discharge simulation

Authors: Peirong Lin; Zong-Liang Yang; David J. Gochis; Wei Yu; David R. Maidment; Marcelo A. Somos-Valenzuela; Cédric H. David

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2018.05.018

URL: https://doi.org/10.1016/j.envsoft.2018.05.018

Publication Date: 2018-06-02

Volume/Issue: Volume 107

References: 44

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Jet Propulsion Laboratory, California Institute of Technology (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

94. A cloud-based flood warning system for forecasting impacts to transportation infrastructure systems

Authors: Mohamed M. Morsy; Jonathan L. Goodall; Gina L. O'Neil; Jeffrey M. Sadler; Daniel Voce; Gamal Hassan; Chris Huxley

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2018.05.007

URL: https://doi.org/10.1016/j.envsoft.2018.05.007

Publication Date: 2018-05-22

Volume/Issue: Volume 107

References: 43

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: Flood forecasting|Extreme Weather|Vulnerability

Funding: Virginia Transportation Research Council

License: N/A

Open Access: N/A

Abstract: N/A

95. Impact of river water levels on the simulation of stream-aquifer exchanges over the Upper Rhine alluvial aquifer (France/Germany)

Authors: Jean-Pierre Vergnes; Florence Habets

Journal: Hydrogeology Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10040-018-1788-0

URL: https://doi.org/10.1007/s10040-018-1788-0

Publication Date: 2018-05-14

Volume/Issue: Volume 26, Issue 7

References: 60

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Abstract: N/A

96. Evolution of river-routing schemes in macro-scale models and their potential for watershed management

Authors: Kashif Shaad (Moore Center for Science, Conservation International, Arlington, Virginia, USA)

Journal: Hydrological Sciences Journal

Publisher: Informa UK Limited

DOI: 10.1080/02626667.2018.1473871

URL: https://doi.org/10.1080/02626667.2018.1473871

Publication Date: 2018-05-08

Volume/Issue: Volume 63, Issue 7

References: 69

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Flora Family Foundation, Betty and Gordon Moore Foundation, Starwood Foundation (and 2 more)

Abstract: N/A

97. Global Estimates of River Flow Wave Travel Times and Implications for Low-Latency Satellite Data

Authors: George H. Allen (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Konstantinos M. Andreadis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Faisal Hossain (University of Washington); James S. Famiglietti (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl077914

URL: https://doi.org/10.1029/2018gl077914

Publication Date: 2018-04-26

Volume/Issue: Volume 45, Issue 15

References: 63

Cited by: 50 publications

Language: en

Subjects: N/A

Keywords: celerity, early flood warning systems, flood wave, SWOT, data latency

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Earth-orbiting satellites provide valuable observations of upstream river conditions worldwide. These observations can be used in real-time applications like early flood warning systems and reservoir operations, provided they are made available to users with sufficient lead time. Yet the temporal requirements for access to satellite-based river data remain uncharacterized for time-sensitive applications. Here we present a global approximation of flow wave travel time to assess the utility of existing and future low-latency/near-real-time satellite products, with an emphasis on the forthcoming SWOT satellite mission. We apply a kinematic wave model to a global hydrography data set and find that global flow waves traveling at their maximum speed take a median travel time of 6, 4, and 3 days to reach their basin terminus, the next downstream city, and the next downstream dam, respectively. Our findings suggest that a recently proposed <=2-day data latency for a low-latency SWOT product is potentially useful for real-time river applications.

98. ORCHIDEE-ROUTING: A new river routing scheme using a high resolution hydrological database

Authors: Trung Nguyen-Quang; Jan Polcher; Agnès Ducharne; Thomas Arsouze; Xudong Zhou; Ana Schneider; Lluís Fita

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2018-57

URL: https://doi.org/10.5194/gmd-2018-57

Publication Date: 2018-04-12

Volume/Issue: N/A

References: N/A

Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Forcing (mathematics)|Flow routing

Funding: École Polytechnique, Université Paris-Saclay

Open Access: N/A

Abstract: . This study presents a revised river routing scheme (RRS) for the Organising Carbon and Hydrology in Dynamic Ecosystems (ORCHIDEE) land surface model. The revision is carried out to benefit from the high resolution topography provided the Hydrological data and maps based on SHuttle Elevation Derivatives at multiple Scales (HydroSHEDS), processed to a resolution of approximately 1 kilometer. The RRS scheme of the ORCHIDEE uses a unit-to-unit routing concept which allows to preserve as much of the hydrological information of the HydroSHEDS as the user requires. The evaluation focuses on 12 rivers of contrasted size and climate which contribute freshwater to the Mediterranean Sea. First, the numerical aspect of the new RRS is investigated, to identify the practical configuration offering the best trade-off between computational cost and simulation quality for ensuing validations. Second, the performance of the revised scheme is evaluated against observations at both monthly and daily timescales. The new RRS captures satisfactorily the seasonal variability of river discharges, although important biases come from the water budget simulated by the ORCHIDEE model. The results highlight that realistic streamflow simulations require accurate precipitation forcing data and a precise river catchment description over a wide range of scales, as permitted by the new RRS. Detailed analyses at the daily timescale show promising performances of this high resolution RRS for replicating river flow variation at various frequencies. Eventually, this RRS is well adapted for further developments in the ORCHIDEE land surface model to assess anthropogenic impacts on river processes (e.g. damming for irrigation operation).

99. Comparison of numerical schemes of river flood routing with an inertial approximation of the Saint Venant equations

Authors: Alice César Fassoni-Andrade (Universidade Federal do Rio Grande do Sul, Brasil); Fernando Mainardi Fan (Universidade Federal do Rio Grande do Sul, Brasil); Walter Collischonn (Universidade Federal do Rio Grande do Sul, Brasil); Artur César Fassoni (Universidade Federal de Itajubá, Brasil); Rodrigo Cauduro Dias de Paiva (Universidade Federal do Rio Grande do Sul, Brasil)

Journal: RBRH

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/2318-0331.0318170069

URL: https://doi.org/10.1590/2318-0331.0318170069

Publication Date: 2018-03-10

Volume/Issue: Volume 23, Issue 0

References: 26

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The one-dimensional flow routing inertial model, formulated as an explicit solution, has advantages over other explicit models used in hydrological models that simplify the Saint-Venant equations. The main advantage is a simple formulation with good results. However, the inertial model is restricted to a small time step to avoid numerical instability. This paper proposes six numerical schemes that modify the one-dimensional inertial model in order to increase the numerical stability of the solution. The proposed numerical schemes were compared to the original scheme in four situations of river's slope (normal, low, high and very high) and in two situations where the river is subject to downstream effects (dam backwater and tides). The results are discussed in terms of stability, peak flow, processing time, volume conservation error and RMSE (Root Mean Square Error). In general, the schemes showed improvement relative to each type of application. In particular, the numerical scheme here called Prog Q(k+1)xQ(k+1) stood out presenting advantages with greater numerical stability in relation to the original scheme. However, this scheme was not successful in the tide simulation situation. In addition, it was observed that the inclusion of the hydraulic radius calculation without simplification in the numerical schemes improved the results without increasing the computational time.

100. An Investigation of Errors in Distributed Models' Stream Discharge Prediction Due to Channel Routing

Authors: Mohamed ElSaadani (Institute for Coastal and Water Research University of Louisiana at Lafayette Lafayette Louisiana USA); Witold F. Krajewski (IIHR--Hydroscience & Engineering University of Iowa Iowa USA); Radoslaw Goska (IIHR--Hydroscience & Engineering University of Iowa Iowa USA); Michael B. Smith (Analysis and Prediction Division National Weather Service Office of Water Prediction Silver Spring Maryland USA)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12627

URL: https://doi.org/10.1111/1752-1688.12627

Publication Date: 2018-02-01

Volume/Issue: Volume 54, Issue 3

References: 22

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: University Corporation for Atmospheric Research

License: N/A

Open Access: N/A

Abstract: Over the summer of 2015, the National Water Center hosted the National Flood Interoperability Experiment (NFIE) Summer Institute. The NFIE organizers introduced a national-scale distributed hydrologic modeling framework that can provide flow estimates at around 2.67 million reaches within the continental United States. The framework generates discharges by coupling a given Land Surface Model (LSM) with the Routing Application for Parallel Computation of Discharge (RAPID). These discharges are then accumulated through the National Hydrography Dataset Plus stream network. The framework can utilize a variety of LSMs to provide the runoff maps to the routing component. The results obtained from this framework suggested that there still exists room for further enhancements to its performance, especially in the area of peak timing and magnitude. The goal of our study was to investigate a single source of the errors in the framework's discharge estimates, which is the routing component. The authors substitute RAPID which is based on the simplified linear Muskingum routing method by the nonlinear routing component the Iowa Flood Center have incorporated in their full hydrologic Hillslope-Link Model. Our results show improvement in model performance across scales due to incorporating new routing methodology.

101. Effects of climate change on streamflow extremes and implications for reservoir inflow in the United States

Authors: Bibi S. Naz; Shih-Chieh Kao; Moetasim Ashfaq; Huilin Gao; Deeksha Rastogi; Sudershan Gangrade

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2017.11.027

URL: https://doi.org/10.1016/j.jhydrol.2017.11.027

Publication Date: 2017-11-16

Volume/Issue: Volume 556

References: 73

Cited by: 78 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Department of Energy (DOE) Water Power Technologies Office, U.S. National Science Foundation, UT-Battelle, LLC (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

102. A Systematic Evaluation of Noah-MP in Simulating Land-Atmosphere Energy, Water, and Carbon Exchanges Over the Continental United States

Authors: Ning Ma (Chinese Academy of Sciences, University of Arizona); Guo-Yue Niu (University of Arizona); Youlong Xia (Environmental Modeling Center, National Centers for Environmental Prediction College Park MD USA); Xitian Cai (Princeton University); Yinsheng Zhang (Chinese Academy of Sciences); Yaoming Ma (Chinese Academy of Sciences); Yuanhao Fang (Hohai University)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jd027597

URL: https://doi.org/10.1002/2017jd027597

Publication Date: 2017-11-08

Volume/Issue: Volume 122, Issue 22

References: 87

Cited by: 105 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: China Postdoctoral Science Foundation, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: Accurate simulation of energy, water, and carbon fluxes exchanging between the land surface and the atmosphere is beneficial for improving terrestrial ecohydrological and climate predictions. We systematically assessed the Noah land surface model (LSM) with mutiparameterization options (Noah-MP) in simulating these fluxes and associated variations in terrestrial water storage (TWS) and snow cover fraction (SCF) against various reference products over 18 United States Geological Survey two-digital hydrological unit code regions of the continental United States (CONUS). In general, Noah-MP captures better the observed seasonal and interregional variability of net radiation, SCF, and runoff than other variables. With a dynamic vegetation model, it overestimates gross primary productivity by 40% and evapotranspiration (ET) by 22% over the whole CONUS domain; however, with a prescribed climatology of leaf area index, it greatly improves ET simulation with relative bias dropping to 4%. It accurately simulates regional TWS dynamics in most regions except those with large lakes or severely affected by irrigation and/or impoundments. Incorporating the lake water storage variations into the modeled TWS variations largely reduces the TWS simulation bias more obviously over the Great Lakes with model efficiency increasing from 0.18 to 0.76. Noah-MP simulates runoff well in most regions except an obvious overestimation (underestimation) in the Rio Grande and Lower Colorado (New England). Compared with North American Land Data Assimilation System Phase 2 (NLDAS-2) LSMs, Noah-MP shows a better ability to simulate runoff and a comparable skill in simulating Rn but a worse skill in simulating ET over most regions. This study suggests that future model developments should focus on improving the representations of vegetation dynamics, lake water storage dynamics, and human activities including irrigation and impoundments.

103. Spatiotemporal Evaluation of Simulated Evapotranspiration and Streamflow over Texas Using the WRF-Hydro-RAPID Modeling Framework

Authors: Peirong Lin (University of Texas at Austin); Mohammad Adnan Rajib (Purdue University); Zong-Liang Yang (Chinese Academy of Sciences, University of Texas at Austin); Marcelo Somos-Valenzuela (University of La Frontera); Venkatesh Merwade (Purdue University); David R. Maidment (University of Texas at Austin); Yan Wang (Hohai University); Li Chen (Hohai University)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12585

URL: https://doi.org/10.1111/1752-1688.12585

Publication Date: 2017-10-23

Volume/Issue: Volume 54, Issue 1

References: 52

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Cynthia and George Mitchell Foundation

License: N/A

Open Access: N/A

Abstract: This study assesses a large-scale hydrologic modeling framework (WRF-Hydro-RAPID) in terms of its high-resolution simulation of evapotranspiration (ET) and streamflow over Texas (drainage area: 464,135 km2). The reference observations used include eight-day ET data from MODIS and FLUXNET, and daily river discharge data from 271 U.S. Geological Survey gauges located across a climate gradient. A recursive digital filter is applied to decompose the river discharge into surface runoff and base flow for comparison with the model counterparts. While the routing component of the model is pre-calibrated, the land component is uncalibrated. Results show the model performance for ET and runoff is aridity-dependent. ET is better predicted in a wet year than in a dry year. Streamflow is better predicted in wet regions with the highest efficiency ~0.7. In comparison, streamflow is most poorly predicted in dry regions with a large positive bias. Modeled ET bias is more strongly correlated with the base flow bias than surface runoff bias. These results complement previous evaluations by incorporating more spatial details. They also help identify potential processes for future model improvements. Indeed, improving the dry region streamflow simulation would require synergistic enhancements of ET, soil moisture and groundwater parameterizations in the current model configuration. Our assessments are important preliminary steps towards accurate large-scale hydrologic forecasts.

104. Towards Real-Time Continental Scale Streamflow Simulation in Continuous and Discrete Space

Authors: Fernando R. Salas (National Water Center National Oceanic and Atmospheric Administration Tuscaloosa Alabama 35401); Marcelo A. Somos-Valenzuela (University of La Frontera); Aubrey Dugger (National Center for Atmospheric Research); David R. Maidment (Center for Research in Water Resources The University of Texas Austin Texas 78712); David J. Gochis (National Center for Atmospheric Research); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Wei Yu (National Center for Atmospheric Research); Deng Ding (Software Development ESRI Redlands California 92373); Edward P. Clark (National Water Center National Oceanic and Atmospheric Administration Tuscaloosa Alabama 35401); Nawajish Noman (Software Development ESRI Redlands California 92373)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12586

URL: https://doi.org/10.1111/1752-1688.12586

Publication Date: 2017-10-16

Volume/Issue: Volume 54, Issue 1

References: 71

Cited by: 77 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Jet Propulsion Laboratory, California Institute of Technology

License: N/A

Open Access: N/A

Abstract: The National Weather Service (NWS) forecasts floods at approximately 3,600 locations across the United States (U.S.). However, the river network, as defined by the 1:100,000 scale National Hydrography Dataset-Plus (NHDPlus) dataset, consists of 2.7 million river segments. Through the National Flood Interoperability Experiment, a continental scale streamflow simulation and forecast system was implemented and continuously operated through the summer of 2015. This system leveraged the WRF-Hydro framework, initialized on a 3-km grid, the Routing Application for the Parallel Computation of Discharge river routing model, operating on the NHDPlus, and real-time atmospheric forcing to continuously forecast streamflow. Although this system produced forecasts, this paper presents a study of the three-month nowcast to demonstrate the capacity to seamlessly predict reach scale streamflow at the continental scale. In addition, this paper evaluates the impact of reservoirs, through a case study in Texas. Validation of the uncalibrated model using observed hourly streamflow at 5,701 U.S. Geological Survey gages shows 26% demonstrate PBias <= |25%|, 11% demonstrate Nash-Sutcliffe Efficiency (NSE) >= 0.25, and 6% demonstrate both PBias <= |25%| and NSE >= 0.25. When evaluating the impact of reservoirs, the analysis shows when reservoirs are included, NSE >= 0.25 for 56% of the gages downstream while NSE >= 0.25 for 11% when they are not. The results presented here provide a benchmark for the evolving hydrology program within the NWS and supports their efforts to develop a reach scale flood forecasting system for the country.

105. Consistent initial conditions for the Saint-Venant equations in river network modeling

Authors: Cheng-Wei Yu; Frank Liu; Ben R. Hodges

Journal: Hydrology and Earth System Sciences

Publisher: Copernicus GmbH

DOI: 10.5194/hess-21-4959-2017

URL: https://doi.org/10.5194/hess-21-4959-2017

Publication Date: 2017-09-29

Volume/Issue: Volume 21, Issue 9

References: 33

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: . Initial conditions for flows and depths (cross-sectional areas) throughout a river network are required for any time-marching (unsteady) solution of the one-dimensional (1-D) hydrodynamic Saint-Venant equations. For a river network modeled with several Strahler orders of tributaries, comprehensive and consistent synoptic data are typically lacking and synthetic starting conditions are needed. Because of underlying nonlinearity, poorly defined or inconsistent initial conditions can lead to convergence problems and long spin-up times in an unsteady solver. Two new approaches are defined and demonstrated herein for computing flows and cross-sectional areas (or depths). These methods can produce an initial condition data set that is consistent with modeled landscape runoff and river geometry boundary conditions at the initial time. These new methods are (1) the pseudo time-marching method (PTM) that iterates toward a steady-state initial condition using an unsteady Saint-Venant solver and (2) the steady-solution method (SSM) that makes use of graph theory for initial flow rates and solution of a steady-state 1-D momentum equation for the channel cross-sectional areas. The PTM is shown to be adequate for short river reaches but is significantly slower and has occasional non-convergent behavior for large river networks. The SSM approach is shown to provide a rapid solution of consistent initial conditions for both small and large networks, albeit with the requirement that additional code must be written rather than applying an existing unsteady Saint-Venant solver.

106. Mapping Outlets of Iowa Flood Center and National Water Center River Networks for Hydrologic Model Comparison

Authors: Felipe Quintero (Iowa Flood Center The University of Iowa C. Maxwell Stanley Hydraulics Laboratory 135 Iowa City Iowa 52242); Witold F. Krajewski (Iowa Flood Center The University of Iowa C. Maxwell Stanley Hydraulics Laboratory 135 Iowa City Iowa 52242)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12554

URL: https://doi.org/10.1111/1752-1688.12554

Publication Date: 2017-08-08

Volume/Issue: Volume 54, Issue 1

References: 27

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: IFC, Consortium of Universities for the Advancement of Hydrologic Sciences (CUAHSI)

License: N/A

Open Access: N/A

Abstract: River networks based on Digital Elevation Model (DEM) data differ depending on the DEM resolution, accuracy, and algorithms used for network extraction. As spatial scale increases, the differences diminish. This study explores methods that identify the scale where networks obtained by different methods agree within some margin of error. The problem is relevant for comparing hydrologic models built around the two networks. An example is the need to compare streamflow prediction from the Hillslope Link Model (HLM) operated by the Iowa Flood Center (IFC) and the National Water Model (NWM) operated by the National Water Center of the National Oceanic and Atmospheric Administration. The HLM uses landscape decomposition into hillslopes and channel links while the NWM uses the NHDPlus dataset as its basic spatial support. While the HLM resolves the scale of the NHDPlus, the outlets of the latter do not necessarily correspond to the nodes of the HLM model. The authors evaluated two methods to map the outlets of NHDPlus to outlets on the IFC network. The methods compare the upstream areas of the channels and their spatial location. Both methods displayed similar performance and identified matches for about 80% of the outlets with a tolerance of 10% in errors in the upstream area. As the aggregation scale increases, the number of matches also increases. At the scale of 100 km2, 90% of the outlets have matches with tolerance of 5%. The authors recommend this scale for comparing the HLM and NWM streamflow predictions.

107. Development of a data model to facilitate rapid watershed delineation

Authors: Scott Haag; Ali Shokoufandeh

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2017.06.009

URL: https://doi.org/10.1016/j.envsoft.2017.06.009

Publication Date: 2017-07-22

Volume/Issue: Volume 122

References: 22

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: William Penn Foundation

License: N/A

Open Access: N/A

Abstract: N/A

108. The critical role of the routing scheme in simulating peak river discharge in global hydrological models

Authors: Fang Zhao; Ted I E Veldkamp; Katja Frieler; Jacob Schewe; Sebastian Ostberg; Sven Willner; Bernhard Schauberger; Simon N Gosling; Hannes Müller Schmied; Felix T Portmann; Guoyong Leng; Maoyi Huang; Xingcai Liu; Qiuhong Tang; Naota Hanasaki; Hester Biemans; Dieter Gerten; Yusuke Satoh; Yadu Pokhrel; Tobias Stacke; Philippe Ciais; Jinfeng Chang; Agnes Ducharne; Matthieu Guimberteau; Yoshihide Wada; Hyungjun Kim; Dai Yamazaki

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/aa7250

URL: https://doi.org/10.1088/1748-9326/aa7250

Publication Date: 2017-06-28

Volume/Issue: Volume 12, Issue 7

References: 80

Cited by: 115 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: German Federal Ministry of Education and Research, European Union Project HELIX, German Federal Ministry of Education and Research (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

109. A Time-Based Framework for Evaluating Hydrologic Routing Methodologies Using Wavelet Transform

Authors: Mohamed ElSaadani; Witold F. Krajewski

Journal: Journal of Water Resource and Protection

Publisher: Scientific Research Publishing, Inc.

DOI: 10.4236/jwarp.2017.97048

URL: https://doi.org/10.4236/jwarp.2017.97048

Publication Date: 2017-06-09

Volume/Issue: Volume 09, Issue 07

References: 20

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: N/A

110. Trade-off between cost and accuracy in large-scale surface water dynamic modeling

Authors: Augusto Getirana (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Christa Peters-Lidard (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Matthew Rodell (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Paul D. Bates (University of Bristol)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017wr020519

URL: https://doi.org/10.1002/2017wr020519

Publication Date: 2017-05-25

Volume/Issue: Volume 53, Issue 6

References: 44

Cited by: 48 publications

Language: en

Subjects: N/A

Keywords: accuracy, kinematic wave equation, trade-off, local inertia formulation, cost (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Recent efforts have led to the development of the local inertia formulation (INER) for an accurate but still cost-efficient representation of surface water dynamics, compared to the widely used kinematic wave equation (KINE). In this study, both formulations are evaluated over the Amazon basin in terms of computational costs and accuracy in simulating streamflows and water levels through synthetic experiments and comparisons against ground-based observations. Varying time steps are considered as part of the evaluation and INER at 60 s time step is adopted as the reference for synthetic experiments. Five hybrid (HYBR) realizations are performed based on maps representing the spatial distribution of the two formulations that physically represent river reach flow dynamics within the domain. Maps have fractions of KINE varying from 35.6 to 82.8%. KINE runs show clear deterioration along the Amazon river and main tributaries, with maximum RMSE values for streamflow and water level reaching 7827 m3 s-1 and 1379 cm near the basin's outlet. However, KINE is at least 25% more efficient than INER with low model sensitivity to longer time steps. A significant improvement is achieved with HYBR, resulting in maximum RMSE values of 3.9-292 m3 s-1 for streamflows and 1.1-28.5 cm for water levels, and cost reduction of 6-16%, depending on the map used. Optimal results using HYBR are obtained when the local inertia formulation is used in about one third of the Amazon basin, reducing computational costs in simulations while preserving accuracy. However, that threshold may vary when applied to different regions, according to their hydrodynamics and geomorphological characteristics.

111. Featured Collection Introduction: National Flood Interoperability Experiment I

Authors: Jim Nelson (Brigham Young University)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12513

URL: https://doi.org/10.1111/1752-1688.12513

Publication Date: 2017-03-06

Volume/Issue: Volume 53, Issue 2

References: 9

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

112. Probabilistic Flood Inundation Forecasting Using Rating Curve Libraries

Authors: Caleb A. Buahin (Utah State University); Nikhil Sangwan (Purdue University); Cassandra Fagan (Department of Civil, Architectural and Environmental Engineering University of Texas Austin Texas 78712); David R. Maidment (Department of Civil, Architectural and Environmental Engineering University of Texas Austin Texas 78712); Jeffery S. Horsburgh (Utah State University); E. James Nelson (Brigham Young University); Venkatesh Merwade (Purdue University); Curtis Rae (Brigham Young University)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12500

URL: https://doi.org/10.1111/1752-1688.12500

Publication Date: 2017-01-27

Volume/Issue: Volume 53, Issue 2

References: 35

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: One approach for performing uncertainty assessment in flood inundation modeling is to use an ensemble of models with different conceptualizations, parameters, and initial and boundary conditions that capture the factors contributing to uncertainty. However, the high computational expense of many hydraulic models renders their use impractical for ensemble forecasting. To address this challenge, we developed a rating curve library method for flood inundation forecasting. This method involves pre-running a hydraulic model using multiple inflows and extracting rating curves, which prescribe a relation between streamflow and stage at various cross sections along a river reach. For a given streamflow, flood stage at each cross section is interpolated from the pre-computed rating curve library to delineate flood inundation depths and extents at a lower computational cost. In this article, we describe the workflow for our rating curve library method and the Rating Curve based Automatic Flood Forecasting (RCAFF) software that automates this workflow. We also investigate the feasibility of using this method to transform ensemble streamflow forecasts into local, probabilistic flood inundation delineations for the Onion and Shoal Creeks in Austin, Texas. While our results show water surface elevations from RCAFF are comparable to those from the hydraulic models, the ensemble streamflow forecasts used as inputs to RCAFF are the largest source of uncertainty in predicting observed floods.

113. Flood Forecasting GIS Water-Flow Visualization Enhancement (WaVE): A Case Study

Authors: Timothy R. Petty; Nawajish Noman; Deng Ding; John B. Gongwer

Journal: Journal of Geographic Information System

Publisher: Scientific Research Publishing, Inc.

DOI: 10.4236/jgis.2016.86055

URL: https://doi.org/10.4236/jgis.2016.86055

Publication Date: 2016-12-22

Volume/Issue: Volume 08, Issue 06

References: 76

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

Abstract: N/A

114. AutoRAPID: A Model for Prompt Streamflow Estimation and Flood Inundation Mapping over Regional to Continental Extents

Authors: Michael L. Follum (Coastal and Hydraulics Laboratory U.S. Army Engineer Research and Development Center 3909 Halls Ferry Road Vicksburg Mississippi 39180); Ahmad A. Tavakoly (Coastal and Hydraulics Laboratory U.S. Army Engineer Research and Development Center 3909 Halls Ferry Road Vicksburg Mississippi 39180); Jeffrey D. Niemann (Colorado State University); Alan D. Snow (Coastal and Hydraulics Laboratory U.S. Army Engineer Research and Development Center 3909 Halls Ferry Road Vicksburg Mississippi 39180)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12476

URL: https://doi.org/10.1111/1752-1688.12476

Publication Date: 2016-10-21

Volume/Issue: Volume 53, Issue 2

References: 53

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Deputy Assistant Secretary of the Army for Research and Technology, Geospatial Intelligence Directorate of the Marine Corps Intelligence Activity

License: N/A

Open Access: N/A

Abstract: This article couples two existing models to quickly generate flow and flood-inundation estimates at high resolutions over large spatial extents for use in emergency response situations. Input data are gridded runoff values from a climate model, which are used by the Routing Application for Parallel computatIon of Discharge (RAPID) model to simulate flow rates within a vector river network. Peak flows in each river reach are then supplied to the AutoRoute model, which produces raster flood inundation maps. The coupled tool (AutoRAPID) is tested for the June 2008 floods in the Midwest and the April-June 2011 floods in the Mississippi Delta. RAPID was implemented from 2005 to 2014 for the entire Mississippi River Basin (1.2 million river reaches) in approximately 45 min. Discretizing a 230,000-km2 area in the Midwest and a 109,500-km2 area in the Mississippi Delta into thirty-nine 1 degrees by 1 degrees tiles, AutoRoute simulated a high-resolution (~10 m) flood inundation map in 20 min for each tile. The hydrographs simulated by RAPID are found to perform better in reaches without influences from unrepresented dams and without backwater effects. Flood inundation maps using the RAPID peak flows vary in accuracy with F-statistic values between 38.1 and 90.9%. Better performance is observed in regions with more accurate peak flows from RAPID and moderate to high topographic relief.

115. Conceptual Framework for the National Flood Interoperability Experiment

Authors: David R. Maidment (University of Texas at Austin)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12474

URL: https://doi.org/10.1111/1752-1688.12474

Publication Date: 2016-10-20

Volume/Issue: Volume 53, Issue 2

References: 16

Cited by: 104 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

Open Access: N/A

Abstract: The National Flood Interoperability Experiment is a research collaboration among academia, National Oceanic and Atmospheric Administration National Weather Service, and government and commercial partners to advance the application of the National Water Model for flood forecasting. In preparation for a Summer Institute at the National Water Center in June-July 2015, a demonstration version of a near real-time, high spatial resolution flood forecasting model was developed for the continental United States. The river and stream network was divided into 2.7 million reaches using the National Hydrography Dataset Plus geospatial dataset and it was demonstrated that the runoff into these stream reaches and the discharge within them could be computed in 10 min at the Texas Advanced Computing Center. This study presents a conceptual framework to connect information from high-resolution flood forecasting with real-time observations and flood inundation mapping and planning for local flood emergency response.

116. Estimation of stream-aquifer exchanges at regional scale using a distributed model: Sensitivity to in-stream water level fluctuations, riverbed elevation and roughness

Authors: Fulvia Baratelli; Nicolas Flipo; Florentina Moatar

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2016.09.041

URL: https://doi.org/10.1016/j.jhydrol.2016.09.041

Publication Date: 2016-09-18

Volume/Issue: Volume 542

References: 74

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European funds, Etablissement Public Loire, Loire River Basin Authority

License: N/A

Open Access: N/A

Abstract: N/A

117. A Comprehensive Python Toolkit for Accessing High-Throughput Computing to Support Large Hydrologic Modeling Tasks

Authors: Scott D. Christensen (Information Technology Laboratory and US Army Engineer Research and Development Center 3909 Halls Ferry Rd Vicksburg Mississippi 39180); Nathan R. Swain (Aquaveo LLC Provo Utah 84604); Norman L. Jones (Brigham Young University); E. James Nelson (Brigham Young University); Alan D. Snow (Coastal and Hydraulics Laboratory US Army Engineer Research and Development Center 3909 Halls Ferry Rd Vicksburg Mississippi 39180); Herman G. Dolder (Aquaveo LLC Provo Utah 84604)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12455

URL: https://doi.org/10.1111/1752-1688.12455

Publication Date: 2016-09-13

Volume/Issue: Volume 53, Issue 2

References: 26

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The National Flood Interoperability Experiment (NFIE) was an undertaking that initiated a transformation in national hydrologic forecasting by providing streamflow forecasts at high spatial resolution over the whole country. This type of large-scale, high-resolution hydrologic modeling requires flexible and scalable tools to handle the resulting computational loads. While high-throughput computing (HTC) and cloud computing provide an ideal resource for large-scale modeling because they are cost-effective and highly scalable, nevertheless, using these tools requires specialized training that is not always common for hydrologists and engineers. In an effort to facilitate the use of HTC resources the National Science Foundation (NSF) funded project, CI-WATER, has developed a set of Python tools that can automate the tasks of provisioning and configuring an HTC environment in the cloud, and creating and submitting jobs to that environment. These tools are packaged into two Python libraries: CondorPy and TethysCluster. Together these libraries provide a comprehensive toolkit for accessing HTC to support hydrologic modeling. Two use cases are described to demonstrate the use of the toolkit, including a web app that was used to support the NFIE national-scale modeling.

118. Continental-Scale River Flow Modeling of the Mississippi River Basin Using High-Resolution NHDPlus Dataset

Authors: Ahmad A. Tavakoly (U.S. Army Engineer Research and Development Center Coastal and Hydraulics Laboratory River Engineering Branch, 3909 Halls Ferry Rd Vicksburg Mississippi 39180); Alan D. Snow (U.S. Army Engineer Research and Development Center Coastal and Hydraulics Laboratory River Engineering Branch, 3909 Halls Ferry Rd Vicksburg Mississippi 39180); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael L. Follum (U.S. Army Engineer Research and Development Center Coastal and Hydraulics Laboratory River Engineering Branch, 3909 Halls Ferry Rd Vicksburg Mississippi 39180); David R. Maidment (University of Texas at Austin); Zong-Liang Yang (University of Texas at Austin)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12456

URL: https://doi.org/10.1111/1752-1688.12456

Publication Date: 2016-09-07

Volume/Issue: Volume 53, Issue 2

References: 64

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Interdisciplinary Science Projects, Deputy Assistant Secretary of the Army for Research and Technology, Jet Propulsion Laboratory, California Institute of Technology

License: N/A

Open Access: N/A

Abstract: As a key component of the National Flood Interoperability Experiment (NFIE), this article presents the continental scale river flow modeling of the Mississippi River Basin (MRB), using high-resolution river data from NHDPlus. The Routing Application for Parallel computatIon of Discharge (RAPID) was applied to the MRB with more than 1.2 million river reaches for a 10-year study (2005-2014). Runoff data from the Variable Infiltration Capacity (VIC) model was used as input to RAPID. This article investigates the effect of topography on RAPID performance, the differences between the VIC-RAPID streamflow simulations in the HUC-2 regions of the MRB, and the impact of major dams on the streamflow simulations. The model performance improved when initial parameter values, especially the Muskingum K parameter, were estimated by taking topography into account. The statistical summary indicates the RAPID model performs better in the Ohio and Tennessee Regions and the Upper and Lower Mississippi River Regions in comparison to the western part of the MRB, due to the better performance of the VIC model. The model accuracy also increases when lakes and reservoirs are considered in the modeling framework. In general, results show the VIC-RAPID streamflow simulation is satisfactory at the continental scale of the MRB.

119. A new open source platform for lowering the barrier for environmental web app development

Authors: Nathan R. Swain; Scott D. Christensen; Alan D. Snow; Herman Dolder; Gonzalo Espinoza-Dávalos; Erfan Goharian; Norman L. Jones; E. James Nelson; Daniel P. Ames; Steven J. Burian

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2016.08.003

URL: https://doi.org/10.1016/j.envsoft.2016.08.003

Publication Date: 2016-08-20

Volume/Issue: Volume 85

References: 69

Cited by: 65 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

120. Open Water Data in Space and Time

Authors: David R. Maidment (Hussein M. Alharthy Centennial Chair in Civil Engineering University of Texas Austin Texas 78712)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12436

URL: https://doi.org/10.1111/1752-1688.12436

Publication Date: 2016-07-05

Volume/Issue: Volume 52, Issue 4

References: 6

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Environmental Systems Research Institute, Microsoft Research

License: N/A

Open Access: N/A

Abstract: An Open Water Data Initiative has been established by the federal government to enhance water information sharing across the United States (U.S.) using standardized web services for geospatial and temporal data. In a parallel effort, the National Weather Service has established a new National Water Center on the Tuscaloosa campus of the University of Alabama, at which a new National Water Model starts operations in June 2016, to continually simulate and forecast streamflow discharge throughout the continental U.S. These two developments support the interoperability of streamflow and hydrologic information in time and space from modeled and observed sources through the use of open standards to share water information.

121. A High-Resolution National-Scale Hydrologic Forecast System from a Global Ensemble Land Surface Model

Authors: Alan D. Snow (Coastal and Hydraulics Laboratory US Army Engineer Research and Development Center 3909 Halls Ferry Rd Vicksburg MS 39180); Scott D. Christensen (Brigham Young University); Nathan R. Swain (Brigham Young University); E. James Nelson (Brigham Young University); Daniel P. Ames (Brigham Young University); Norman L. Jones (Brigham Young University); Deng Ding (Esri Redlands CA 92373); Nawajish S. Noman (Esri Redlands CA 92373); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Florian Pappenberger (European Centre for Medium-Range Weather Forecasts (ECMWF)); Ervin Zsoter (European Centre for Medium-Range Weather Forecasts (ECMWF))

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12434

URL: https://doi.org/10.1111/1752-1688.12434

Publication Date: 2016-06-27

Volume/Issue: Volume 52, Issue 4

References: 31

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

Open Access: N/A

Abstract: Warning systems with the ability to predict floods several days in advance have the potential to benefit tens of millions of people. Accordingly, large-scale streamflow prediction systems such as the Advanced Hydrologic Prediction Service or the Global Flood Awareness System are limited to coarse resolutions. This article presents a method for routing global runoff ensemble forecasts and global historical runoff generated by the European Centre for Medium-Range Weather Forecasts model using the Routing Application for Parallel computatIon of Discharge to produce high spatial resolution 15-day stream forecasts, approximate recurrence intervals, and warning points at locations where streamflow is predicted to exceed the recurrence interval thresholds. The processing method involves distributing the computations using computer clusters to facilitate processing of large watersheds with high-density stream networks. In addition, the Streamflow Prediction Tool web application was developed for visualizing analyzed results at both the regional level and at the reach level of high-density stream networks. The application formed part of the base hydrologic forecasting service available to the National Flood Interoperability Experiment and can potentially transform the nation's forecast ability by incorporating ensemble predictions at the nearly 2.7 million reaches of the National Hydrography Plus Version 2 Dataset into the national forecasting system.

122. A decade of RAPID--Reflections on the development of an open source geoscience code

Authors: Cédric H. David (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); James S. Famiglietti (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zong-Liang Yang (University of Texas at Austin); Florence Habets (UMR 7619 METIS, CNRS, UPMC Paris France); David R. Maidment (University of Texas at Austin)

Journal: Earth and Space Science

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015ea000142

URL: https://doi.org/10.1002/2015ea000142

Publication Date: 2016-04-10

Volume/Issue: Volume 3, Issue 5

References: 68

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: open, data, RAPID, code, license

Funding: Microsoft Research

Abstract: Earth science increasingly relies on computer-based methods and many government agencies now require further sharing of the digital products they helped fund. Earth scientists, while often supportive of more transparency in the methods they develop, are concerned by this recent requirement and puzzled by its multiple implications. This paper therefore presents a reflection on the numerous aspects of sharing code and data in the general field of computer modeling of dynamic Earth processes. Our reflection is based on 10 years of development of an open source model called the Routing Application for Parallel Computation of Discharge (RAPID) that simulates the propagation of water flow waves in river networks. Three consecutive but distinct phases of the sharing process are highlighted here: opening, exposing, and consolidating. Each one of these phases is presented as an independent and tractable increment aligned with the various stages of code development and justified based on the size of the users community. Several aspects of digital scholarship are presented here including licenses, documentation, websites, citable code and data repositories, and testing. While the many existing services facilitate the sharing of digital research products, digital scholarship also raises community challenges related to technical training, self-perceived inadequacy, community contribution, acknowledgment and performance assessment, and sustainable sharing.

123. An Analysis of Water Data Systems to Inform the Open Water Data Initiative

Authors: David Blodgett (Office of Water Information U.S. Geological Survey 8505 Research Way Middleton Wisconsin 53562); Emily Read (Office of Water Information U.S. Geological Survey 8505 Research Way Middleton Wisconsin 53562); Jessica Lucido (Office of Water Information U.S. Geological Survey 8505 Research Way Middleton Wisconsin 53562); Tad Slawecki (LimnoTech Ann Arbor Michigan 48108); Dwane Young (Office of Water U.S. Environmental Protection Agency Washington DC 20460)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12417

URL: https://doi.org/10.1111/1752-1688.12417

Publication Date: 2016-04-10

Volume/Issue: Volume 52, Issue 4

References: 12

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: Improving access to data and fostering open exchange of water information is foundational to solving water resources issues. In this vein, the Department of the Interior's Assistant Secretary for Water and Science put forward the charge to undertake an Open Water Data Initiative (OWDI) that would prioritize and accelerate work toward better water data infrastructure. The goal of the OWDI is to build out the Open Water Web (OWW). We therefore considered the OWW in terms of four conceptual functions: water data cataloging, water data as a service, enriching water data, and community for water data. To describe the current state of the OWW and identify areas needing improvement, we conducted an analysis of existing systems using a standard model for describing distributed systems and their business requirements. Our analysis considered three OWDI-focused use cases--flooding, drought, and contaminant transport--and then examined the landscape of other existing applications that support the Open Water Web. The analysis, which includes a discussion of observed successful practices of cataloging, serving, enriching, and building community around water resources data, demonstrates that we have made significant progress toward the needed infrastructure, although challenges remain. The further development of the OWW can be greatly informed by the interpretation and findings of our analysis.

124. From Global to Local: Providing Actionable Flood Forecast Information in a Cloud-Based Computing Environment

Authors: J. Fidel Perez (Brigham Young University); Nathan R. Swain (Brigham Young University); Herman G. Dolder (Brigham Young University); Scott D. Christensen (Brigham Young University); Alan D. Snow (Brigham Young University); E. James Nelson (Brigham Young University); Norman L. Jones (Brigham Young University)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12392

URL: https://doi.org/10.1111/1752-1688.12392

Publication Date: 2016-02-11

Volume/Issue: Volume 52, Issue 4

References: 54

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: Hydrological modelling

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Global and continental scale flood forecast provide coarse resolution flood forecast, but from the perspective of emergency management, flood warnings should be detailed and specific to local conditions. The desired refinement can be provided by the use of downscaling global scale models and through the use of distributed hydrologic models to produce a high-resolution flood forecast. Three major challenges associated with transforming global flood forecasting to a local scale are addressed in this work. The first is using open-source software tools to provide access to multiple data sources and lowering the barriers for users in management agencies at local level. This can be done through the Tethys Platform that enables web water resources modeling applications. The second is finding a practical solution for the computational requirements associated with running complex models and performing multiple simulations. This is done using Tethys Cluster that manages distributed and cloud computing resources as a companion to the Tethys Platform for web app development. The third challenge is discovering ways to downscale the forecasts from the global extent to the local context. Three modeling strategies have been tested to address this, including downscaling of coarse resolution global runoff models to high-resolution stream networks and routing with Routing Application for Parallel computatIon of Discharge (RAPID), the use of hierarchical Gridded Surface and Subsurface Hydrologic Analysis (GSSHA) distributed models, and pre-computed distributed GSSHA models.

125. Development and evaluation of a physically-based lake level model for water resource management: A case study for Lake Buchanan, Texas

Authors: Peirong Lin; Zong-Liang Yang; Xitian Cai; Cédric H. David

Journal: Journal of Hydrology: Regional Studies

Publisher: Elsevier BV

DOI: 10.1016/j.ejrh.2015.08.005

URL: https://doi.org/10.1016/j.ejrh.2015.08.005

Publication Date: 2015-12-01

Volume/Issue: Volume 4

References: 43

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

Open Access: N/A

Abstract: N/A

126. A GIS Framework for Regional Modeling of Riverine Nitrogen Transport: Case Study, San Antonio and Guadalupe Basins

Authors: Ahmad A. Tavakoly (Coastal and Hydraulics Laboratory River Engineering Branch U.S. Army Corps of Engineers 3909 Halls Ferry Rd Vicksburg Mississippi 39180); David R. Maidment (University of Texas at Austin); James W. McClelland (University of Texas at Austin); Tim Whiteaker (University of Texas at Austin); Zong-Liang Yang (University of Texas at Austin); Claire Griffin (University of Texas at Austin); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lisa Meyer (Hilcorp Energy Company Houston Texas 77002)

Journal: JAWRA Journal of the American Water Resources Association

Publisher: Wiley

DOI: 10.1111/1752-1688.12355

URL: https://doi.org/10.1111/1752-1688.12355

Publication Date: 2015-09-24

Volume/Issue: Volume 52, Issue 1

References: 52

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Interdisciplinary Science Project, USACE Coastal and Hydraulics Laboratory

License: N/A

Open Access: N/A

Abstract: This article presents a framework for integrating a regional geographic information system (GIS)-based nitrogen dataset (Texas Anthropogenic Nitrogen Dataset, TX-ANB) and a GIS-based river routing model (Routing Application for Parallel computation of Discharge) to simulate steady-state riverine total nitrogen (TN) transport in river networks containing thousands of reaches. A two-year case study was conducted in the San Antonio and Guadalupe basins during dry and wet years (2008 and 2009, respectively). This article investigates TN export in urbanized (San Antonio) vs. rural (Guadalupe) drainage basins and considers the effect of reservoirs on TN transport. Simulated TN export values are within 10 percent of measured export values for selected stations in 2008 and 2009. Results show that in both years the San Antonio basin contributed a larger quantity than the Guadalupe basin of delivered TN to the coastal ocean. The San Antonio basin is affected by urban activities including point sources, associated with the city of San Antonio, in addition to greater agricultural activities. The Guadalupe basin lacks major metropolitan areas and is dominated by rangeland, rather than fertilized agricultural fields. Both basins delivered more TN to coastal waters in 2009 than in 2008. Furthermore, TN removal in the San Antonio and Guadalupe basins is inversely related to stream orders: the higher the order the more TN delivery (or the less TN removal).

127. Enhanced fixed-size parallel speedup with the Muskingum method using a trans-boundary approach and a large subbasins approximation

Authors: Cédric H. David (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); James S. Famiglietti (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zong-Liang Yang (University of Texas at Austin); Victor Eijkhout (University of Texas at Austin)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014wr016650

URL: https://doi.org/10.1002/2014wr016650

Publication Date: 2015-06-24

Volume/Issue: Volume 51, Issue 9

References: 73

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: parallel, speedup, Muskingum, trans-boundary, network (and 1 more)

Funding: Jet Propulsion Laboratory, University of California

License: N/A

Open Access: N/A

Abstract: N/A

128. Evaluation of Regional-Scale River Depth Simulations Using Various Routing Schemes within a Hydrometeorological Modeling Framework for the Preparation of the SWOT Mission

Authors: Vincent Häfliger (+ Centre National d'Études Spatiales, Toulouse, France, * CNRM-GAME, UMR 3589, Météo-France, CNRS, Toulouse, France); Eric Martin (* CNRM-GAME, UMR 3589, Météo-France, CNRS, Toulouse, France); Aaron Boone (* CNRM-GAME, UMR 3589, Météo-France, CNRS, Toulouse, France); Florence Habets (# UMR 7619 METIS, CNRS, UPMC, Paris, France); Cédric H. David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Pierre-A. Garambois (& Université de Toulouse, INPT, UPS, Institut de Mécanique des Fluides de Toulouse, Toulouse, France); Hélène Roux (& Université de Toulouse, INPT, UPS, Institut de Mécanique des Fluides de Toulouse, Toulouse, France); Sophie Ricci (** CERFACS-URA 1875, Toulouse, France); Lucie Berthon (** CERFACS-URA 1875, Toulouse, France); Anthony Thévenin (** CERFACS-URA 1875, Toulouse, France); Sylvain Biancamaria (Institut de Recherche pour le Développement (IRD))

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/jhm-d-14-0107.1

URL: https://doi.org/10.1175/jhm-d-14-0107.1

Publication Date: 2015-06-11

Volume/Issue: Volume 16, Issue 4

References: 58

Cited by: 9 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The Surface Water and Ocean Topography (SWOT) mission will provide free water surface elevations, slopes, and river widths for rivers wider than 50 m. Models must be prepared to use this new finescale information by explicitly simulating the link between runoff and the river channel hydraulics. This study assesses one regional hydrometeorological model's ability to simulate river depths. The Garonne catchment in southwestern France (56 000 km2) has been chosen for the availability of operational gauges in the river network and finescale hydraulic models over two reaches of the river. Several routing schemes, ranging from the simple Muskingum method to time-variable parameter kinematic and diffusive waves schemes, are tested. The results show that the variable flow velocity schemes are advantageous for discharge computations when compared to the original Muskingum routing method. Additionally, comparisons between river depth computations and in situ observations in the downstream Garonne River led to root-mean-square errors of 50-60 cm in the improved Muskingum method and 40-50 cm in the kinematic-diffusive wave method. The results also highlight SWOT's potential to improve the characterization of hydrological processes for subbasins larger than 10 000 km2, the importance of an accurate digital elevation model, and the need for

spatially varying hydraulic parameters.

129. Development and implementation of river-routing process module in a regional climate model and its evaluation in Korean river basins

Authors: Ji-Woo Lee (University of California Los Angeles); Song-You Hong (Korea Institute of Atmospheric Prediction System Seoul South Korea); Jung-Eun Esther Kim (Colorado State University, Earth System Research Laboratory National Oceanic and Atmospheric Administration Boulder Colorado USA); Kei Yoshimura (University of Tokyo); Suryun Ham (University of Tokyo); Minsu Joh (Korea Institute of Science and Technology Information Daejeon South Korea)

Journal: Journal of Geophysical Research: Atmospheres

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jd022698

URL: https://doi.org/10.1002/2014jd022698

Publication Date: 2015-04-09

Volume/Issue: Volume 120, Issue 10

References: 81

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Korea Institute of Science and Technology Information (KISTI)

License: N/A

Open Access: N/A

Abstract: This study assessed the potential for river discharge simulation by implementing an online river-routing scheme into the regional climate model (RCM) framework as a unified subroutine module and investigated the sensitivity of simulated river flows in response to changes in spatial resolutions in RCM and river-routing scheme. The river-routing scheme gathers runoff from the RCM and advects them horizontally along the river drainage network. The dynamical downscaling simulations were driven by reanalysis at the boundaries for the period of 2000-2010, using different grid sizes for RCM (50 and 12.5 km) and for river-routing scheme (0.5 degrees, 0.25 degrees, and 0.125 degrees). Simulated river discharge was evaluated throughout the three largest river basins in Korea. The simulation results showed potential for river discharge modeling in the RCM framework. The model generally captured the seasonal and monthly variabilities, and the daily scale peaks. From the resolution sensitivity experiments, it was confirmed that high-resolution RCM enhances the reproducibility of river discharge; however, the lack of sophistication of the current river-routing scheme, which was originally developed for continental and macroscale application, mitigates taking advantage of enhanced resolution in river model. On the basis of our findings and experiences in this study, we revealed several considerable issues for future developments of river simulation in the RCM framework.

130. Applying microprocessor analysis methods to river network modelling

Authors: Frank Liu; Ben R. Hodges

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2013.09.013

URL: https://doi.org/10.1016/j.envsoft.2013.09.013

Publication Date: 2013-10-17

Volume/Issue: Volume 52

References: 68

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

131. Improving computational efficiency in global river models by implementing the local inertial flow equation and a vector-based river network map

Authors: Dai Yamazaki (University of Bristol); Gustavo A. M. de Almeida (University of Southampton, University of Bristol); Paul D. Bates (University of Bristol)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/wrcr.20552

URL: https://doi.org/10.1002/wrcr.20552

Publication Date: 2013-10-12

Volume/Issue: Volume 49, Issue 11

References: 53

Cited by: 176 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Global river models are an essential tool for both earth system studies and water resources assessments. As advanced physical processes have been implemented in global river models, increasing computational cost has become problematic for executing ensemble or long-term simulations. To improve computational efficiency, we here propose the use of a local inertial flow equation combined with a vector-based river network map. A local inertial equation, a simplified formulation of the shallow water equations, was introduced to replace a diffusion wave equation. A vector-based river network map which flexibly discretizes river segments was adopted in order to replace the traditional grid-based map which is based on a Cartesian grid coordinate system. The computational efficiency of the proposed flow routing and river network map was tested by executing hydrodynamic simulations with the CaMa-Flood global river model. The simulation results suggest that the computational efficiency can be improved by more than 300% by applying the local inertial equation. It can be improved by a further 60% by implementing the vector-based river network map instead of a grid-based map. It is found that the vector-based map with evenly distributed flow distances between calculation units allows longer time steps compared to the grid-based map because the latter has very short flow distances between calculation units at high latitudes which critically limit time step length. Considering the improvement in simulation speed, the local inertial equation, and a vector-based river network map are preferable in global hydrodynamic simulations with high computational demands such as ensemble or long-term experiments.

132. Quantification of the upstream-to-downstream influence in the Muskingum method and implications for speedup in parallel computations of river flow

Authors: Cédric H. David (University of California Irvine, University of Texas at Austin); Zong-Liang Yang (University of Texas at Austin); James S. Famiglietti (University of California Irvine)

Journal: Water Resources Research

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/wrcr.20250

URL: https://doi.org/10.1002/wrcr.20250

Publication Date: 2013-04-17

Volume/Issue: Volume 49, Issue 5

References: 28

Cited by: 25 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Interdisciplinary Science Project, U.S. National Aeronautics and Space Administration, University of California Office of the President Multicampus Research Programs (and 1 more)

License: N/A

Open Access: N/A

Abstract: Key PointsThe influence between two river reaches in a Muskingum update is quantified.This influence decreases with distance and becomes null in computer addition.Such allows new parallel computing methods that are unprecedentedly fast.

133. Regional-scale river flow modeling using off-the-shelf runoff products, thousands of mapped rivers and hundreds of stream flow gauges

Authors: Cédric H. David; Zong-Liang Yang; Seungbum Hong

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2012.12.011

URL: https://doi.org/10.1016/j.envsoft.2012.12.011

Publication Date: 2013-02-06

Volume/Issue: Volume 42

References: 45

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

134. River Network Routing on the NHDPlus Dataset

Authors: Cédric H. David (University of Texas at Austin); David R. Maidment (University of Texas at Austin); Guo-Yue Niu (University of Arizona, University of Texas at Austin); Zong-Liang Yang (University of Texas at Austin); Florence Habets (@ UMR-Sisyphe 7619, CNRS, UPMC, MINES ParisTech, Paris, France); Victor Eijkhout (University of Texas at Austin)

Journal: Journal of Hydrometeorology

Publisher: American Meteorological Society

DOI: 10.1175/2011jhm1345.1

URL: https://doi.org/10.1175/2011jhm1345.1

Publication Date: 2011-04-19

Volume/Issue: Volume 12, Issue 5

References: 48

Cited by: 176 publications

Language: N/A

Subjects: N/A

Keywords: North America, Streamflow, Rivers, Numerical analysis/modeling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The mapped rivers and streams of the contiguous United States are available in a geographic information system (GIS) dataset called National Hydrography Dataset Plus (NHDPlus). This hydrographic dataset has about 3 million river and water body reaches along with information on how they are connected into networks. The U.S. Geological Survey (USGS) National Water Information System (NWIS) provides streamflow observations at about 20 thousand gauges located on the NHDPlus river network. A river network model called Routing Application for Parallel Computation of Discharge (RAPID) is developed for the NHDPlus river network whose lateral inflow to the river network is calculated by a land surface model. A matrix-based version of the Muskingum method is developed herein, which RAPID uses to calculate flow and volume of water in all reaches of a river network with many thousands of reaches, including at ungauged locations. Gauges situated across river basins (not only at basin outlets) are used to automatically optimize the Muskingum parameters and to assess river flow computations, hence allowing the diagnosis of runoff computations provided by land surface models. RAPID is applied to the Guadalupe and San Antonio River basins in Texas, where flow wave celerities are estimated at multiple locations using 15-min data and can be reproduced reasonably with RAPID. This river model can be adapted for parallel computing and although the matrix method initially adds a large overhead, river flow results can be obtained faster than with the traditional Muskingum method when using a few processing cores, as demonstrated in a synthetic study using the upper Mississippi River basin.

135. RAPID applied to the SIM-France model

Authors: Cédric H. David; Florence Habets; David R. Maidment; Zong-Liang Yang

Journal: Hydrological Processes

Publisher: Wiley

DOI: 10.1002/hyp.8070

URL: https://doi.org/10.1002/hyp.8070

Publication Date: 2011-03-02

Volume/Issue: Volume 25, Issue 22

References: 31

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

136. Modeling the impact of in-stream water level fluctuations on stream-aquifer interactions at the regional scale

Authors: Firas Saleh; Nicolas Flipo; Florence Habets; Agnès Ducharne; Ludovic Oudin; Pascal Viennot; Emmanuel Ledoux

Journal: Journal of Hydrology

Publisher: Elsevier BV

DOI: 10.1016/j.jhydrol.2011.02.001

URL: https://doi.org/10.1016/j.jhydrol.2011.02.001

Publication Date: 2011-02-09

Volume/Issue: Volume 400, Issue 3-4

References: 70

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A